# Unsupervised Learning & Feature Reduction on UCI HAR

Dataset: *Human Activity Recognition Using Smartphones* (UCI) — 561 engineered features computed from 2.56 s windows of smartphone accelerometer/gyroscope signals, 30 subjects, 6 activities.

**Protocol.** The provided training set is split by subject into **train** and **dev**; all exploration, reduction and model selection use only these. The provided **test** set is loaded and scaled in Phase 1 (with the scaler fitted on train only) and cached, but it is not analysed, plotted or scored until the single final evaluation in Phase 5. Activity labels are never given to the unsupervised methods — they are only used afterwards to evaluate them.

All figures are saved to `figures/` (PNG, dpi=200), all numeric results to `results/` (CSV), and the processed arrays to `cache/`.

## Phase 1 - Setup

The cells below import the libraries, fix `random_state=42`, and define the shared helpers (locating and loading the dataset, saving figures/tables, the subject-wise split, the cache). They then load the provided train and test sets, split the provided training set into train/dev **by subject**, fit a `StandardScaler` on train only, transform train/dev/test with it, and cache the processed arrays to `cache/har_processed.npz` so later phases load them quickly.

In [1]:
```
import time
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import display
from sklearn.model_selection import GroupShuffleSplit
from sklearn.preprocessing import StandardScaler

RANDOM_STATE = 42
np.random.seed(RANDOM_STATE)

PROJECT_DIR = Path.cwd()
FIG_DIR = PROJECT_DIR / "figures"
```

```
RES_DIR = PROJECT_DIR / "results"
CACHE_DIR = PROJECT_DIR / "cache"
CACHE_FILE = CACHE_DIR / "har_processed.npz"
for d in (FIG_DIR, RES_DIR, CACHE_DIR):
    d.mkdir(exist_ok=True)

sns.set_theme(style="whitegrid", context="notebook")
pd.set_option("display.max_columns", 30)
pd.set_option("display.width", 140)
```

In [2]:

```
# ---------- Shared helpers: locating and loading the dataset ----------

def find_data_dir(root=PROJECT_DIR):
    """Find the extracted 'UCI HAR Dataset' directory (has train/, test/, features.txt, activity_labels.txt).

    Searches the project folder; ignores macOS '__MACOSX' metadata copies. Nothing is downloaded or extracted.
    """
    for features_file in sorted(root.rglob("features.txt")):
        d = features_file.parent
        if "__MACOSX" in d.parts:
            continue
        if (d / "activity_labels.txt").exists() and (d / "train").is_dir() and (d / "test").is_dir():
            return d
    raise FileNotFoundError(f"No 'UCI HAR Dataset' directory found under {root}")


DATA_DIR = find_data_dir()
print("Dataset directory:", DATA_DIR.relative_to(PROJECT_DIR))


def load_activity_labels():
    """Map integer activity code -> activity name."""
    df = pd.read_csv(DATA_DIR / "activity_labels.txt", sep=r"\s+", header=None, names=["code", "activity"])
    return dict(zip(df["code"], df["activity"]))


def load_feature_names():
    """Return the 561 feature names, made unique (features.txt repeats some band-energy names)."""
    names = pd.read_csv(DATA_DIR / "features.txt", sep=r"\s+", header=None, names=["idx", "name"])["name"]
    counts, unique = {}, []
    for n in names:
        counts[n] = counts.get(n, 0) + 1
```

```
        unique.append(n if counts[n] == 1 else f"{n}_{counts[n]}")
    return unique


ACTIVITY_LABELS = load_activity_labels()
ACTIVITY_ORDER = [ACTIVITY_LABELS[k] for k in sorted(ACTIVITY_LABELS)]
FEATURE_NAMES = load_feature_names()


def load_split(split):
    """Load one provided split ('train' or 'test').

    Returns X (DataFrame, 561 named features), y (Series of activity NAMES), subject (int Series).
    """
    assert split in ("train", "test")
    d = DATA_DIR / split
    X = pd.read_csv(d / f"X_{split}.txt", sep=r"\s+", header=None, names=FEATURE_NAMES)
    codes = pd.read_csv(d / f"y_{split}.txt", header=None)[0]
    y = pd.Series(pd.Categorical(codes.map(ACTIVITY_LABELS), categories=ACTIVITY_ORDER), name="activity")
    subject = pd.read_csv(d / f"subject_{split}.txt", header=None)[0].rename("subject")
    return X, y, subject


# ---------- Shared helpers: saving outputs ----------

def save_fig(fig, name):
    """Save a figure to figures/<name>.png at dpi=200 (it is also shown inline)."""
    path = FIG_DIR / f"{name}.png"
    fig.savefig(path, dpi=200, bbox_inches="tight")
    print(f"saved {path.relative_to(PROJECT_DIR)}")


def save_table(df, name, index=True):
    """Save a table to results/<name>.csv."""
    path = RES_DIR / f"{name}.csv"
    df.to_csv(path, index=index)
    print(f"saved {path.relative_to(PROJECT_DIR)}")


TIMINGS = []

def record_timing(phase, step, seconds):
```

```
    """Log a wall-clock timing to results/timings.csv. Timings vary run to run, so they live in this one
    file instead of the per-phase result tables (which then stay identical across re-runs)."""
    TIMINGS.append({"phase": phase, "step": step, "seconds": round(seconds, 3)})
    pd.DataFrame(TIMINGS).to_csv(RES_DIR / "timings.csv", index=False)


# ---------- Shared helpers: splitting ----------

def subject_split(X, y, subject, dev_size=0.2):
    """Split by SUBJECT with GroupShuffleSplit (~dev_size of subjects to dev) so no person is in both."""
    gss = GroupShuffleSplit(n_splits=1, test_size=dev_size, random_state=RANDOM_STATE)
    tr_idx, dev_idx = next(gss.split(X, y, groups=subject))
    take = lambda obj, idx: obj.iloc[idx].reset_index(drop=True)
    return (take(X, tr_idx), take(y, tr_idx), take(subject, tr_idx),
            take(X, dev_idx), take(y, dev_idx), take(subject, dev_idx))


# ---------- Shared helpers: processed-data cache ----------

def load_cache(path=CACHE_FILE):
    """Load the processed arrays written in Phase 1.

    Returns a dict with X_{train,dev,test} (standardized float arrays), y_{train,dev,test} (activity names),
    subject_{train,dev,test}, feature_names, activity_order, scaler_mean, scaler_scale.
    """
    with np.load(path, allow_pickle=False) as z:
        return {k: z[k] for k in z.files}
```

```
Dataset directory: human+activity+recognition+using+smartphones\UCI HAR Dataset\UCI HAR Dataset
```

In [3]:

```
# Load the provided train and test sets
t0 = time.perf_counter()
X_train_full, y_train_full, subject_train_full = load_split("train")
X_test, y_test, subject_test = load_split("test")
print(f"Loaded in {time.perf_counter() - t0:.1f}s")
print(f"Provided train: X={X_train_full.shape}, {subject_train_full.nunique()} subjects, "
      f"{X_train_full.isna().sum().sum()} missing values")
print(f"Provided test:  X={X_test.shape}, {subject_test.nunique()} subjects, "
      f"{X_test.isna().sum().sum()} missing values")
print(f"Features: {len(FEATURE_NAMES)} ({len(set(FEATURE_NAMES))} unique after de-duplication); "
      f"activities: {ACTIVITY_ORDER}")
X_train_full.iloc[:5, :8]
```

```
Loaded in 2.0s
Provided train: X=(7352, 561), 21 subjects, 0 missing values
Provided test:  X=(2947, 561), 9 subjects, 0 missing values
Features: 561 (561 unique after de-duplication); activities: ['WALKING', 'WALKING_UPSTAIRS', 'WALKING_DOWNSTAIRS', 'SITTING', 'STA
NDING', 'LAYING']
```

Out[3]:

| | tBodyAcc-mean()-X | tBodyAcc-mean()-Y | tBodyAcc-mean()-Z | tBodyAcc-std()-X | tBodyAcc-std()-Y | tBodyAcc-std()-Z | tBodyAcc-mad()-X | tBodyAcc-mad()-Y |
|---|---|---|---|---|---|---|---|---|
| **0** | 0.288585 | -0.020294 | -0.132905 | -0.995279 | -0.983111 | -0.913526 | -0.995112 | -0.983185 |
| **1** | 0.278419 | -0.016411 | -0.123520 | -0.998245 | -0.975300 | -0.960322 | -0.998807 | -0.974914 |
| **2** | 0.279653 | -0.019467 | -0.113462 | -0.995380 | -0.967187 | -0.978944 | -0.996520 | -0.963668 |
| **3** | 0.279174 | -0.026201 | -0.123283 | -0.996091 | -0.983403 | -0.990675 | -0.997099 | -0.982750 |
| **4** | 0.276629 | -0.016570 | -0.115362 | -0.998139 | -0.980817 | -0.990482 | -0.998321 | -0.979672 |

In [4]:

```python
# Split the provided training set into train / dev BY SUBJECT
X_train, y_train, subject_train, X_dev, y_dev, subject_dev = subject_split(
    X_train_full, y_train_full, subject_train_full, dev_size=0.2)

assert set(subject_train).isdisjoint(subject_dev), "subject leakage between train and dev"
assert set(subject_train_full).isdisjoint(subject_test), "provided train/test share subjects"

for name, s in [("train", subject_train), ("dev", subject_dev), ("test", subject_test)]:
    print(f"{name:5s}: {len(s):5d} rows, {s.nunique():2d} subjects -> {sorted(s.unique().tolist())}")

split_summary = pd.DataFrame({
    "train": y_train.value_counts().reindex(ACTIVITY_ORDER),
    "dev": y_dev.value_counts().reindex(ACTIVITY_ORDER),
    "test": y_test.value_counts().reindex(ACTIVITY_ORDER),
})
split_summary.index.name = "activity"
split_summary.loc["TOTAL"] = split_summary.sum()
for c in ["train", "dev", "test"]:
    split_summary[f"{c}_%"] = (100 * split_summary[c] / split_summary.loc["TOTAL", c]).round(1)
split_summary.loc["n_subjects", ["train", "dev", "test"]] = [
    subject_train.nunique(), subject_dev.nunique(), subject_test.nunique()]
split_summary[["train", "dev", "test"]] = split_summary[["train", "dev", "test"]].astype(int)

save_table(split_summary, "phase1_split_summary")
display(split_summary)
```

```
train:  5551 rows, 16 subjects -> [5, 6, 7, 8, 11, 14, 16, 17, 19, 21, 22, 23, 26, 28, 29, 30]
dev  :  1801 rows,  5 subjects -> [1, 3, 15, 25, 27]
test :  2947 rows,  9 subjects -> [2, 4, 9, 10, 12, 13, 18, 20, 24]
saved results\phase1_split_summary.csv
```

| activity | train | dev | test | train_% | dev_% | test_% |
|---|---|---|---|---|---|---|
| **WALKING** | 888 | 338 | 496 | 16.0 | 18.8 | 16.8 |
| **WALKING_UPSTAIRS** | 797 | 276 | 471 | 14.4 | 15.3 | 16.0 |
| **WALKING_DOWNSTAIRS** | 744 | 242 | 420 | 13.4 | 13.4 | 14.3 |
| **SITTING** | 993 | 293 | 491 | 17.9 | 16.3 | 16.7 |
| **STANDING** | 1053 | 321 | 532 | 19.0 | 17.8 | 18.1 |
| **LAYING** | 1076 | 331 | 537 | 19.4 | 18.4 | 18.2 |
| **TOTAL** | 5551 | 1801 | 2947 | 100.0 | 100.0 | 100.0 |
| **n_subjects** | 16 | 5 | 9 | NaN | NaN | NaN |

In [5]:
```
# Fit StandardScaler on TRAIN ONLY, then transform train, dev and test with it
scaler = StandardScaler().fit(X_train)
Xs_train = scaler.transform(X_train)
Xs_dev = scaler.transform(X_dev)
Xs_test = scaler.transform(X_test)

print(f"train after scaling: mean={Xs_train.mean():+.2e}, std={Xs_train.std():.3f}")
print(f"dev   after scaling: mean={Xs_dev.mean():+.3f}, std={Xs_dev.std():.3f}  (not exactly 0/1: scaler fit on train)")
print(f"Zero-variance features in train: {(scaler.scale_ == 0).sum()}")
```

```
train after scaling: mean=+4.06e-19, std=1.000
dev   after scaling: mean=-0.048, std=0.901  (not exactly 0/1: scaler fit on train)
Zero-variance features in train: 0
```

In [6]:
```
# Cache the processed arrays so later phases load them quickly
t0 = time.perf_counter()
np.savez_compressed(
    CACHE_FILE,
    X_train=Xs_train.astype(np.float32), X_dev=Xs_dev.astype(np.float32), X_test=Xs_test.astype(np.float32),
    y_train=y_train.to_numpy(dtype=str), y_dev=y_dev.to_numpy(dtype=str), y_test=y_test.to_numpy(dtype=str),
    subject_train=subject_train.to_numpy(), subject_dev=subject_dev.to_numpy(), subject_test=subject_test.to_numpy(),
    feature_names=np.array(FEATURE_NAMES), activity_order=np.array(ACTIVITY_ORDER),
    scaler_mean=scaler.mean_, scaler_scale=scaler.scale_,
)
print(f"Wrote {CACHE_FILE.relative_to(PROJECT_DIR)} ({CACHE_FILE.stat().st_size / 1e6:.1f} MB) "
      f"in {time.perf_counter() - t0:.1f}s")

# Round-trip check
t0 = time.perf_counter()
data = load_cache()
print(f"Reloaded in {time.perf_counter() - t0:.2f}s:",
```

```
      {k: v.shape for k, v in data.items() if k.startswith(("X_", "y_", "subject_"))})
assert np.allclose(data["X_train"], Xs_train, atol=1e-5) and (data["y_test"] == y_test.to_numpy(dtype=str)).all()

# Free every test-set object: the test set is reloaded from the cache only for the final evaluation in Phase 5
del X_test, y_test, subject_test, Xs_test, data
```

```
Wrote cache\har_processed.npz (19.7 MB) in 1.0s
Reloaded in 0.12s: {'X_train': (5551, 561), 'X_dev': (1801, 561), 'X_test': (2947, 561), 'y_train': (5551,), 'y_dev': (1801,), 'y_
test': (2947,), 'subject_train': (5551,), 'subject_dev': (1801,), 'subject_test': (2947,)}
```

# Phase 2 - EDA

The cells below load the cached standardized arrays and use the **train split only**. They plot the activity counts, fit PCA with all components on train, plot the explained variance ratio of the first 50 components and the cumulative explained variance, locate the elbow programmatically (maximum distance to the line joining the first and last points of the ratio curve over the first 50 components — the kneedle idea), report how many components reach 90% and 95% cumulative variance, and show the train data on PC1 vs PC2 coloured by activity.

In [7]:
```
# Load the cached arrays from Phase 1 and keep the train split only (test arrays are dropped immediately;
# the test set is loaded again only in the final evaluation of Phase 5)
data = {k: v for k, v in load_cache().items() if not k.endswith("_test")}
X_tr = data["X_train"]
y_tr = pd.Categorical(data["y_train"], categories=ACTIVITY_ORDER)

# One fixed colour per activity, reused in every later figure
ACTIVITY_PALETTE = dict(zip(ACTIVITY_ORDER, sns.color_palette("tab10", len(ACTIVITY_ORDER))))

print(f"train: X={X_tr.shape} (standardized with train statistics)")
```

```
train: X=(5551, 561) (standardized with train statistics)
```

## 2.1 Activity counts (train)

In [8]:
```
train_counts = pd.Series(y_tr).value_counts().reindex(ACTIVITY_ORDER).rename("n_windows").to_frame()
train_counts["pct"] = (100 * train_counts["n_windows"] / train_counts["n_windows"].sum()).round(1)
train_counts.index.name = "activity"
save_table(train_counts, "phase2_train_activity_counts")
display(train_counts)

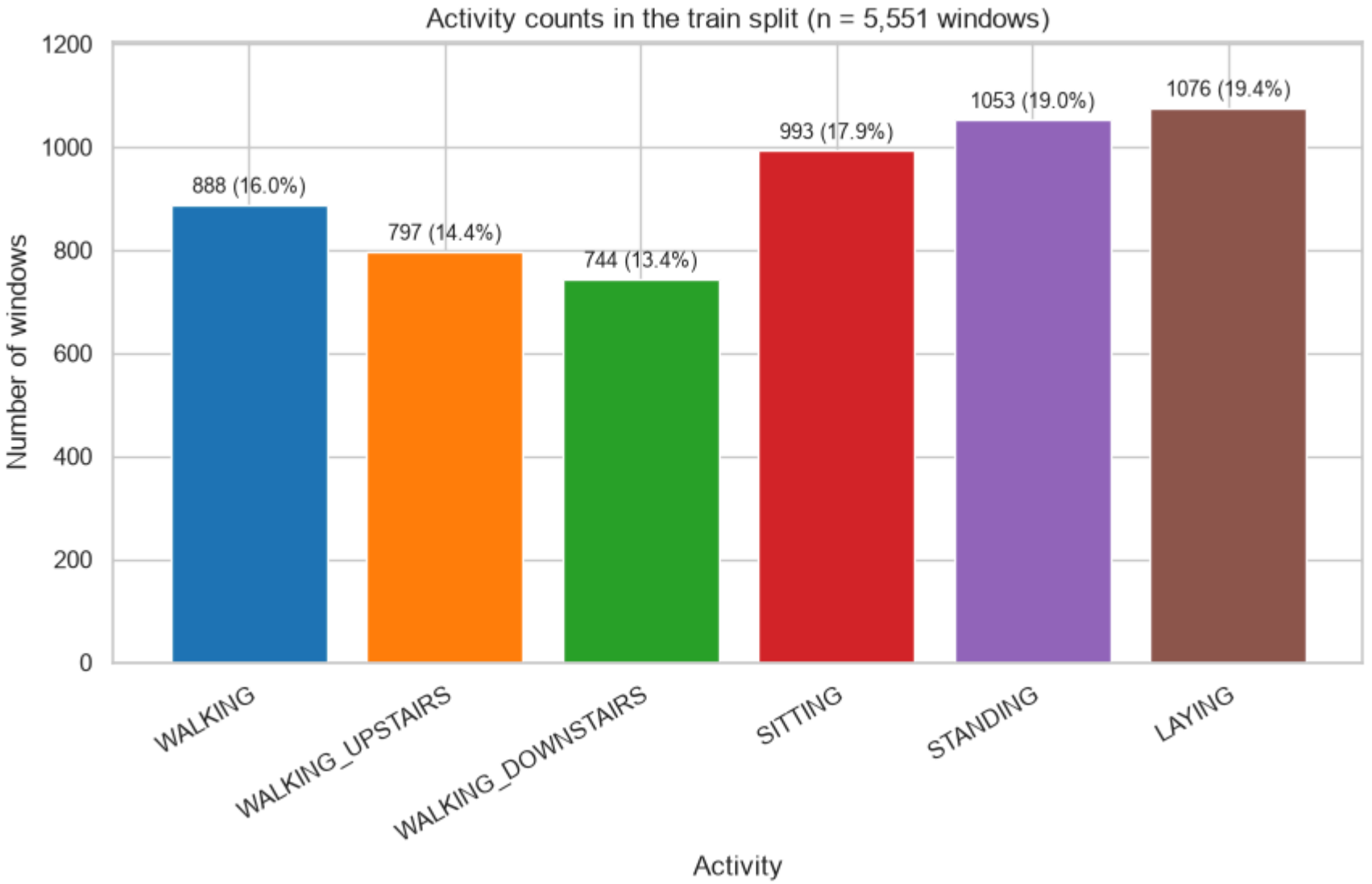
fig, ax = plt.subplots(figsize=(10, 5))
```

```
bars = ax.bar(train_counts.index, train_counts["n_windows"],
              color=[ACTIVITY_PALETTE[a] for a in train_counts.index])
ax.bar_label(bars, labels=[f"{n} ({p}%)" for n, p in zip(train_counts["n_windows"], train_counts["pct"])],
             padding=3, fontsize=9)
ax.set_title(f"Activity counts in the train split (n = {len(y_tr):,} windows)")
ax.set_xlabel("Activity")
ax.set_ylabel("Number of windows")
plt.setp(ax.get_xticklabels(), rotation=30, ha="right", rotation_mode="anchor")
ax.set_ylim(0, train_counts["n_windows"].max() * 1.12)
save_fig(fig, "phase2_train_activity_counts")
plt.show()
```

```
saved results\phase2_train_activity_counts.csv
```

| activity | n_windows | pct |
|---|---|---|
| **WALKING** | 888 | 16.0 |
| **WALKING_UPSTAIRS** | 797 | 14.4 |
| **WALKING_DOWNSTAIRS** | 744 | 13.4 |
| **SITTING** | 993 | 17.9 |
| **STANDING** | 1053 | 19.0 |
| **LAYING** | 1076 | 19.4 |

```
saved figures\phase2_train_activity_counts.png
```




## 2.2 PCA on train (all components)

In [9]:
```
from sklearn.decomposition import PCA

t0 = time.perf_counter()
pca = PCA(n_components=None, random_state=RANDOM_STATE).fit(X_tr)
pca_fit_seconds = time.perf_counter() - t0
```

```
record_timing("phase2", "pca_fit_all_components", pca_fit_seconds)

evr = pca.explained_variance_ratio_
cum_evr = np.cumsum(evr)
print(f"PCA fitted on train in {pca_fit_seconds:.2f}s: {pca.n_components_} components")

pca_variance = pd.DataFrame({
    "component": np.arange(1, len(evr) + 1),
    "explained_variance_ratio": evr,
    "cumulative_explained_variance": cum_evr,
})
save_table(pca_variance, "phase2_pca_explained_variance", index=False)
display(pca_variance.head(10).round(4))
```

```
PCA fitted on train in 0.18s: 561 components
saved results\phase2_pca_explained_variance.csv
```

| | component | explained_variance_ratio | cumulative_explained_variance |
|---|---|---|---|
| **0** | 1 | 0.5110 | 0.5110 |
| **1** | 2 | 0.0626 | 0.5736 |
| **2** | 3 | 0.0280 | 0.6016 |
| **3** | 4 | 0.0259 | 0.6275 |
| **4** | 5 | 0.0197 | 0.6472 |
| **5** | 6 | 0.0182 | 0.6654 |
| **6** | 7 | 0.0141 | 0.6795 |
| **7** | 8 | 0.0118 | 0.6912 |
| **8** | 9 | 0.0106 | 0.7018 |
| **9** | 10 | 0.0100 | 0.7119 |

In [10]:

```
def elbow_max_distance(y):
    """Elbow of a curve = point with maximum perpendicular distance to the straight line joining its first
    and last points (the kneedle idea). Both axes are min-max normalised to [0, 1] first so the result does
    not depend on their scales. Returns the 1-based index of the elbow and the distance curve."""
    y = np.asarray(y, dtype=float)
    x = np.arange(1, len(y) + 1, dtype=float)
    xn = (x - x[0]) / (x[-1] - x[0])
    yn = (y - y.min()) / (y.max() - y.min())
    p1, p2 = np.array([xn[0], yn[0]]), np.array([xn[-1], yn[-1]])
    dist = np.abs((p2[0] - p1[0]) * (p1[1] - yn) - (p1[0] - xn) * (p2[1] - p1[1])) / np.linalg.norm(p2 - p1)
    return int(x[np.argmax(dist)]), dist


N_PLOT = 50
```

```
k_elbow, elbow_dist = elbow_max_distance(evr[:N_PLOT])
k90 = int(np.searchsorted(cum_evr, 0.90) + 1)
k95 = int(np.searchsorted(cum_evr, 0.95) + 1)

print(f"Elbow (max distance to line, first {N_PLOT} components): k = {k_elbow} "
      f"-> cumulative variance {cum_evr[k_elbow - 1]:.1%}")
print(f"Components to reach 90% cumulative variance: {k90}")
print(f"Components to reach 95% cumulative variance: {k95}")

pca_summary = pd.DataFrame({
    "quantity": [f"elbow_k (max-distance, first {N_PLOT} comps)", "k_90pct", "k_95pct", "n_components_total"],
    "value": [k_elbow, k90, k95, pca.n_components_],
    "cumulative_variance_at_k": [cum_evr[k_elbow - 1], cum_evr[k90 - 1], cum_evr[k95 - 1], 1.0],
})
save_table(pca_summary, "phase2_pca_summary", index=False)
display(pca_summary.round(4))
```

```
Elbow (max distance to line, first 50 components): k = 3 -> cumulative variance 60.2%
Components to reach 90% cumulative variance: 61
Components to reach 95% cumulative variance: 100
saved results\phase2_pca_summary.csv
```

| | quantity | value | cumulative_variance_at_k |
|---|---|---|---|
| **0** | elbow_k (max-distance, first 50 comps) | 3 | 0.6016 |
| **1** | k_90pct | 61 | 0.9003 |
| **2** | k_95pct | 100 | 0.9507 |
| **3** | n_components_total | 561 | 1.0000 |

In [11]:
```
# Robustness of the elbow to the cutoff: recompute on the first 30, 50 and 100 components.
# The official k stays the 50-component value (k_elbow above).
elbow_cutoffs = pd.DataFrame([
    {"cutoff_n_components": n, "elbow_k": elbow_max_distance(evr[:n])[0]} for n in (30, 50, 100)
])
elbow_cutoffs["cumulative_variance_at_k"] = cum_evr[elbow_cutoffs["elbow_k"] - 1]
elbow_cutoffs["official"] = elbow_cutoffs["cutoff_n_components"] == N_PLOT

for row in elbow_cutoffs.itertuples():
    tag = "  <- official k" if row.official else ""
    print(f"Elbow on first {row.cutoff_n_components:3d} components: k = {row.elbow_k} "
          f"({row.cumulative_variance_at_k:.1%} cumulative){tag}")
same = elbow_cutoffs["elbow_k"].nunique() == 1
print("k is identical across cutoffs" if same else "k changes with the cutoff -- see table")
```

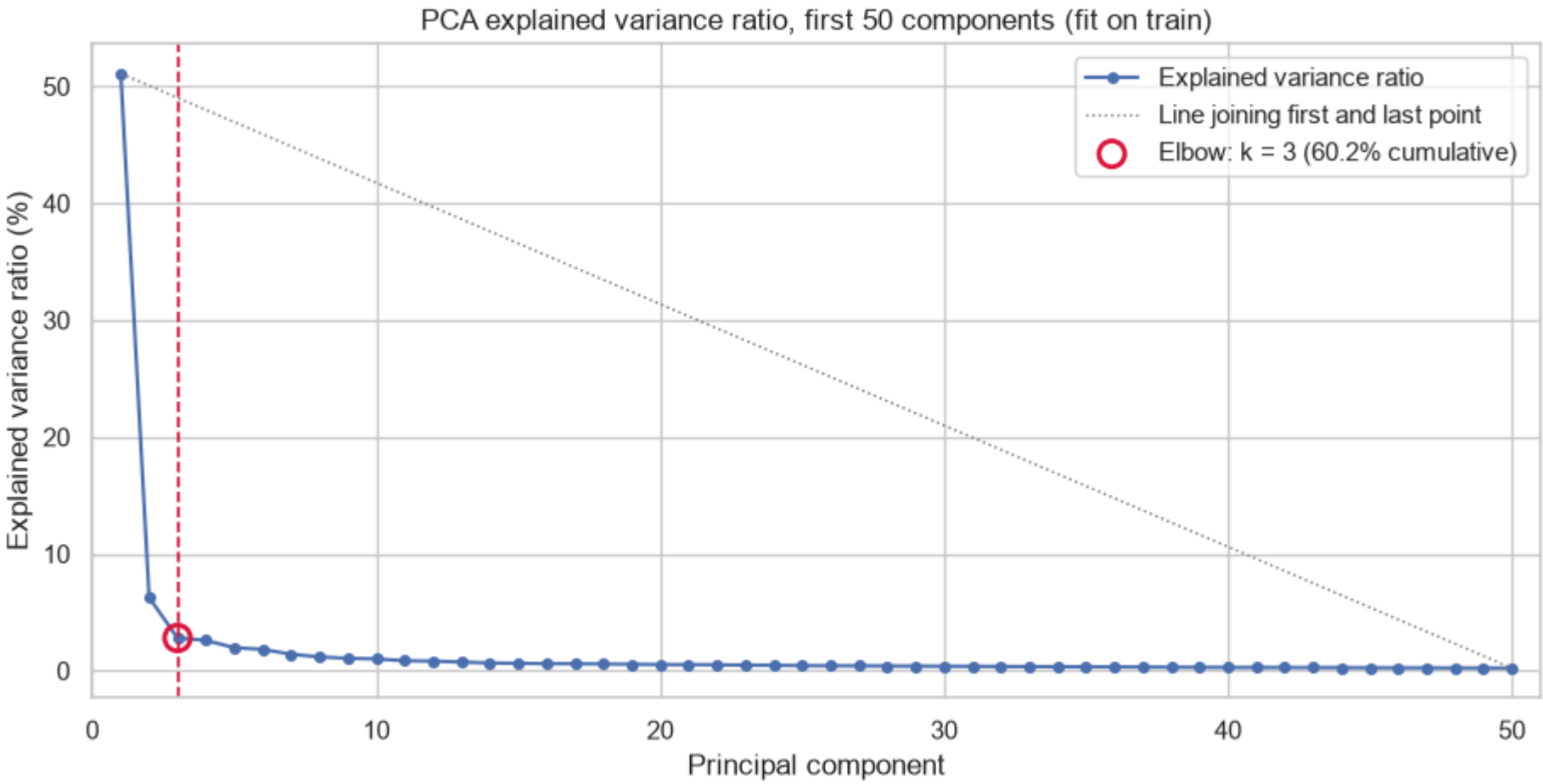
```
save_table(elbow_cutoffs, "phase2_elbow_cutoff_sensitivity", index=False)
display(elbow_cutoffs.round(4))
```

```
Elbow on first  30 components: k = 3 (60.2% cumulative)
Elbow on first  50 components: k = 3 (60.2% cumulative)  <- official k
Elbow on first 100 components: k = 3 (60.2% cumulative)
k is identical across cutoffs
saved results\phase2_elbow_cutoff_sensitivity.csv
```

| | cutoff_n_components | elbow_k | cumulative_variance_at_k | official |
|---|---|---|---|---|
| **0** | 30 | 3 | 0.6016 | False |
| **1** | 50 | 3 | 0.6016 | True |
| **2** | 100 | 3 | 0.6016 | False |

In [12]:
```
# Explained variance ratio, first 50 components, with the elbow marked
comps = np.arange(1, N_PLOT + 1)
fig, ax = plt.subplots(figsize=(11, 5))
ax.plot(comps, 100 * evr[:N_PLOT], marker="o", ms=4, lw=1.5, color="#4C72B0", label="Explained variance ratio")
ax.plot([1, N_PLOT], 100 * evr[[0, N_PLOT - 1]], ls=":", color="gray", lw=1, label="Line joining first and last point")
ax.axvline(k_elbow, color="crimson", ls="--", lw=1.2)
ax.scatter([k_elbow], [100 * evr[k_elbow - 1]], s=120, facecolor="none", edgecolor="crimson", lw=2, zorder=5,
           label=f"Elbow: k = {k_elbow} ({100 * cum_evr[k_elbow - 1]:.1f}% cumulative)")
ax.set_title(f"PCA explained variance ratio, first {N_PLOT} components (fit on train)")
ax.set_xlabel("Principal component")
ax.set_ylabel("Explained variance ratio (%)")
ax.set_xlim(0, N_PLOT + 1)
ax.legend()
save_fig(fig, "phase2_pca_explained_variance")
plt.show()
```

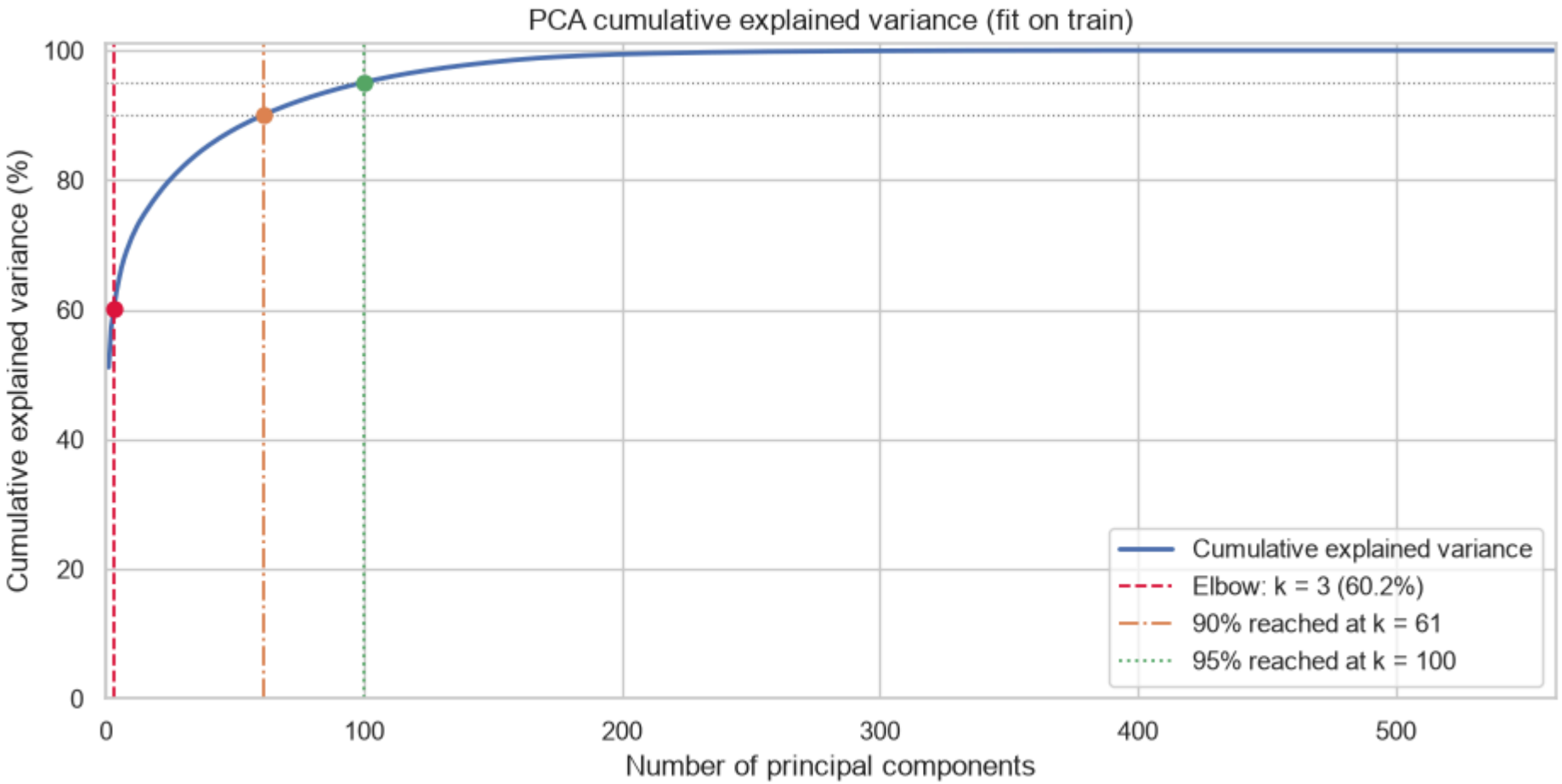
```
saved figures\phase2_pca_explained_variance.png
```




In [13]:
```python
# Cumulative explained variance (all components) with elbow, 90% and 95% marked
all_comps = np.arange(1, len(cum_evr) + 1)
fig, ax = plt.subplots(figsize=(11, 5))
ax.plot(all_comps, 100 * cum_evr, lw=2, color="#4C72B0", label="Cumulative explained variance")
markers = [(k_elbow, "crimson", "--", f"Elbow: k = {k_elbow} ({100 * cum_evr[k_elbow - 1]:.1f}%)"),
           (k90, "#DD8452", "-.", f"90% reached at k = {k90}"),
           (k95, "#55A868", ":", f"95% reached at k = {k95}")]
for k, color, ls, label in markers:
    ax.axvline(k, color=color, ls=ls, lw=1.3, label=label)
    ax.scatter([k], [100 * cum_evr[k - 1]], color=color, s=40, zorder=5)
for level in (90, 95):
    ax.axhline(level, color="gray", lw=0.8, ls=":")
ax.set_title("PCA cumulative explained variance (fit on train)")
```

```
ax.set_xlabel("Number of principal components")
ax.set_ylabel("Cumulative explained variance (%)")
ax.set_xlim(0, len(cum_evr) + 1)
ax.set_ylim(0, 101)
ax.legend(loc="lower right")
save_fig(fig, "phase2_pca_cumulative_variance")
plt.show()
```

```
saved figures\phase2_pca_cumulative_variance.png
```

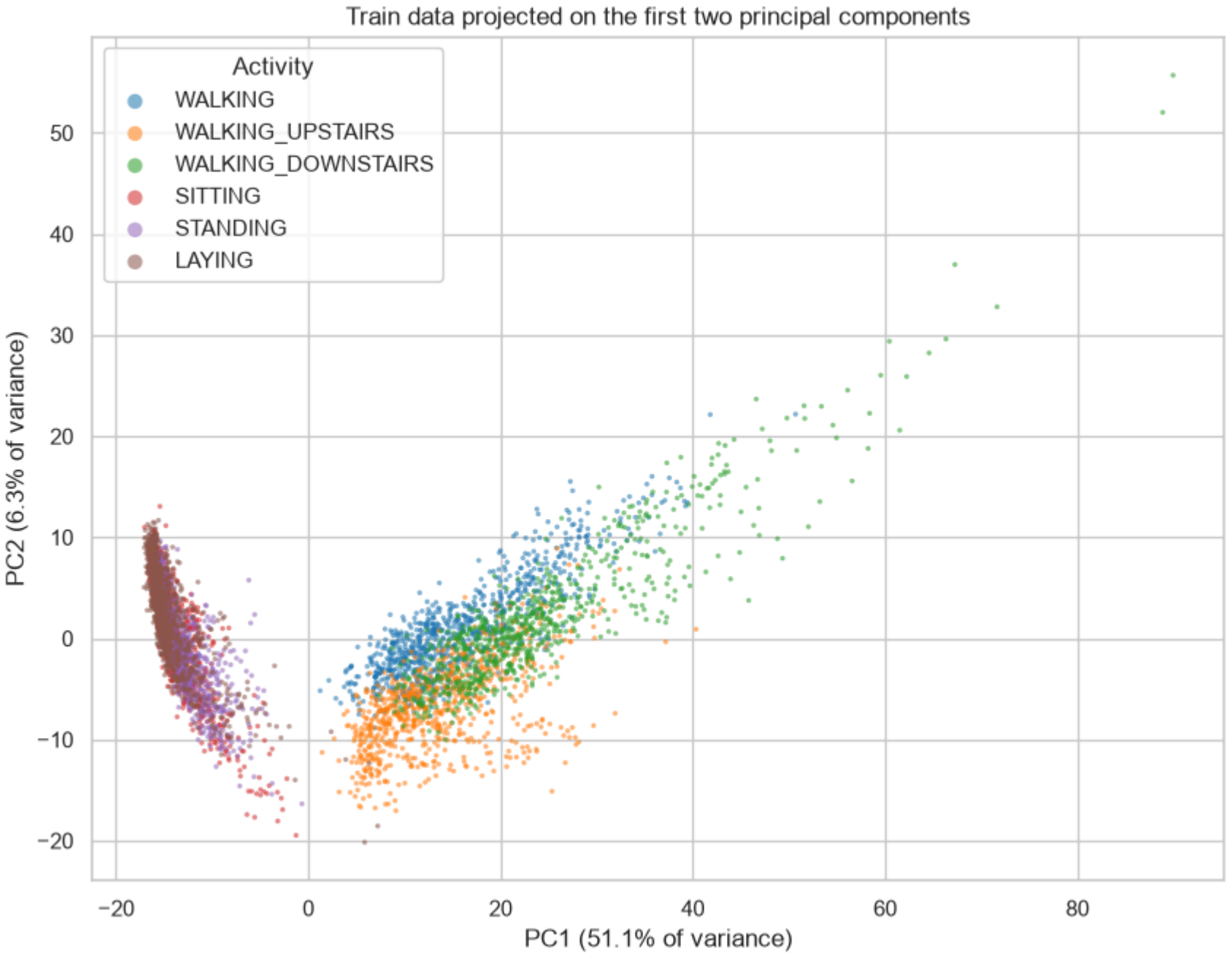



## 2.3 Train data on PC1 vs PC2

In [14]:
```
Z_tr = pca.transform(X_tr)[:, :2]

fig, ax = plt.subplots(figsize=(10, 7.5))
for act in ACTIVITY_ORDER:
    m = np.asarray(y_tr) == act
    ax.scatter(Z_tr[m, 0], Z_tr[m, 1], s=6, alpha=0.55, color=ACTIVITY_PALETTE[act], label=act, edgecolors="none")
ax.set_title("Train data projected on the first two principal components")
ax.set_xlabel(f"PC1 ({100 * evr[0]:.1f}% of variance)")
ax.set_ylabel(f"PC2 ({100 * evr[1]:.1f}% of variance)")
ax.legend(title="Activity", markerscale=3, framealpha=0.9)
save_fig(fig, "phase2_pca_pc1_pc2_scatter")
plt.show()
```

```
saved figures\phase2_pca_pc1_pc2_scatter.png
```



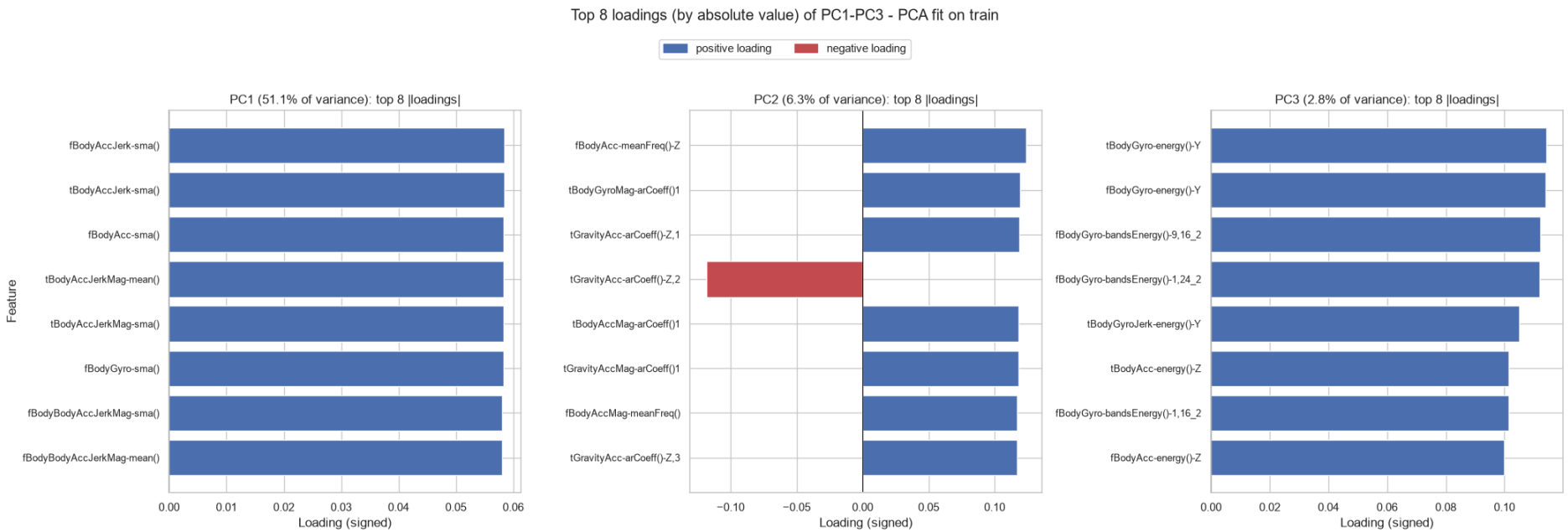

## 2.4 What PC1, PC2 and PC3 represent: top loadings

Each panel shows the 8 features with the largest absolute loading (coefficient in the principal axis, on the standardized features) for one component. Bars keep their sign: features on the same side move together along that component, features on opposite sides move in opposite directions.

In [15]:
```
N_LOAD = 8
loading_rows = []
fig, axes = plt.subplots(1, 3, figsize=(21, 6.5))
for i, ax in enumerate(axes):
    comp = pd.Series(pca.components_[i], index=FEATURE_NAMES)
    top_load = comp.reindex(comp.abs().sort_values(ascending=False).index[:N_LOAD])
    colors = np.where(top_load.values >= 0, "#4C72B0", "#C44E52")
    ax.barh(top_load.index, top_load.values, color=colors)
    ax.invert_yaxis()
    ax.axvline(0, color="black", lw=0.8)
    ax.set_title(f"PC{i + 1} ({100 * evr[i]:.1f}% of variance): top {N_LOAD} |loadings|")
    ax.set_xlabel("Loading (signed)")
    ax.set_ylabel("Feature" if i == 0 else "")
    ax.tick_params(axis="y", labelsize=9)
    loading_rows += [{"component": f"PC{i + 1}", "rank": r + 1, "feature": f, "loading": v}
                     for r, (f, v) in enumerate(top_load.items())]
from matplotlib.patches import Patch
fig.legend(handles=[Patch(color="#4C72B0", label="positive loading"), Patch(color="#C44E52", label="negative loading")],
           loc="upper center", ncol=2, bbox_to_anchor=(0.5, 1.04))
fig.suptitle(f"Top {N_LOAD} loadings (by absolute value) of PC1-PC3 - PCA fit on train", y=1.09)
fig.tight_layout()
save_fig(fig, "phase2_pca_top_loadings")
plt.show()

pca_top_loadings = pd.DataFrame(loading_rows)
save_table(pca_top_loadings, "phase2_pca_top_loadings", index=False)
display(pca_top_loadings.pivot(index="rank", columns="component", values="feature"))
```

```
saved figures\phase2_pca_top_loadings.png
```




saved `results\phase2_pca_top_loadings.csv`

| component | PC1 | PC2 | PC3 |
|---|---|---|---|
| rank | | | |
| 1 | fBodyAccJerk-sma() | fBodyAcc-meanFreq()-Z | tBodyGyro-energy()-Y |
| 2 | tBodyAccJerk-sma() | tBodyGyroMag-arCoeff()1 | fBodyGyro-energy()-Y |
| 3 | fBodyAcc-sma() | tGravityAcc-arCoeff()-Z,1 | fBodyGyro-bandsEnergy()-9,16_2 |
| 4 | tBodyAccJerkMag-mean() | tGravityAcc-arCoeff()-Z,2 | fBodyGyro-bandsEnergy()-1,24_2 |
| 5 | tBodyAccJerkMag-sma() | tBodyAccMag-arCoeff()1 | tBodyGyroJerk-energy()-Y |
| 6 | fBodyGyro-sma() | tGravityAccMag-arCoeff()1 | tBodyAcc-energy()-Z |
| 7 | fBodyBodyAccJerkMag-sma() | fBodyAccMag-meanFreq() | fBodyGyro-bandsEnergy()-1,16_2 |
| 8 | fBodyBodyAccJerkMag-mean() | tGravityAcc-arCoeff()-Z,3 | fBodyAcc-energy()-Z |

**What each component represents** (from the loadings above):

- **PC1 - overall movement intensity:** all top loadings are positive and nearly equal (~0.058) magnitude/area features ( `sma` , `Mag-mean` ) of body acceleration, jerk and gyroscope, so PC1 is high for walking and low for stationary windows.
- **PC2 - temporal/spectral shape rather than amplitude:** dominated by autoregressive coefficients ( `arCoeff` ) and mean frequency ( `meanFreq` ) of body and gravity acceleration, i.e. how the signal oscillates, not how strong it is.
- **PC3 - rotation around the phone's Y axis:** gyroscope Y-axis energy (incl. `bandsEnergy ..._2` , the Y block) plus Z-axis body-acceleration energy.

## Report - 1. Exploratory Analysis

**Setup.** The provided training set (7,352 windows, 21 subjects) was split **by subject** into train (5,551 windows, 16 subjects) and dev (1,801 windows, 5 subjects) with `GroupShuffleSplit`, so no person appears in both. The provided test set (2,947 windows, 9 different subjects) was scaled with the train-fitted `StandardScaler` and cached in Phase 1, but no model was trained or evaluated on it until the final evaluation in Section 5.6.

**Activity distribution** (`phase2_train_activity_counts.png`). The six activities are reasonably balanced in train, from 13.4% (walking downstairs) to 19.4% (laying). Static activities are slightly more common than moving ones. No class is rare, so accuracy is a meaningful metric and no resampling was needed.

**Explained variance and choice of k** (`phase2_pca_explained_variance.png`, `phase2_pca_cumulative_variance.png`). Variance is extremely concentrated: PC1 alone explains 51.1%, PC2 6.3%, PC3 2.8%, and every component after that adds less than 3%. We located the elbow with the max-distance-to-line method on the explained variance ratio of the first 50 components, which gives **k = 3** (60.2% cumulative variance). The result is identical when the curve is cut at 30 or 100 components, so k does not depend on the cutoff. A visual reading could argue for a softer bend near k ≈ 9, where each component falls to about 1% of the variance. The cumulative curve shows the cost of a small k: 90% of the variance needs 61 components and 95% needs 100. Most of the information is spread across a long tail, a tension we test directly in Section 3.

**What the components mean** (`phase2_pca_top_loadings.png`). PC1's top loadings are all signal-magnitude-area and jerk features (e.g. `fBodyAccJerk-sma()`, `tBodyAccJerkMag-mean()`), so PC1 measures **overall motion intensity**. PC2 is dominated by autoregressive coefficients and mean frequency, describing the **rhythm/frequency content** of the signal. PC3 is dominated by gyroscope energy on the Y axis, i.e. **rotation**.

**2D projection** (`phase2_pca_pc1_pc2_scatter.png`). The data splits into two clearly separated groups along PC1: static activities (sitting, standing, laying) packed into one dense blob on the left, and moving activities on the right. Within the moving group, walking upstairs separates partly toward negative PC2, while walking and walking downstairs overlap heavily. Walking downstairs forms the high-intensity tail. Sitting, standing and laying are almost indistinguishable in these two dimensions. This picture predicts most of what follows: moving vs stationary is easy, separating the three static postures is hard.

# Phase 3 - Clustering

The cells below cluster the **train split** without using the activity labels; the labels are only used afterwards to evaluate (contingency tables and Adjusted Rand Index, ARI).

- **K-Means** on all 561 standardized features for k = 2..10, choosing k by silhouette score ( `sample_size=3000, random_state=42` ), then comparing the silhouette-chosen k with k = 6 (the true number of activities).
- **DBSCAN** on the first `k_elbow` PCA components from Phase 2. A k-distance plot ( `min_samples=10` ) locates the knee, eps is swept around it, and the best eps is the one with the highest silhouette among settings with at least 2 clusters and at most 10% noise (an unsupervised criterion; ARI is reported, not used to choose).

In [16]:
```
from sklearn.cluster import KMeans, DBSCAN
from sklearn.metrics import silhouette_score, adjusted_rand_score
from sklearn.neighbors import NearestNeighbors

SIL_KW = dict(sample_size=3000, random_state=RANDOM_STATE)
y_tr_arr = np.asarray(y_tr)
Z2_tr = pca.transform(X_tr)[:, :2]          # PC1/PC2 for plotting (PCA fitted on train in Phase 2)


def contingency(clusters, labels, noise_label=-1):
    """Cluster x activity count table; clusters ordered by their dominant activity, noise last."""
    names = np.where(clusters == noise_label, "noise", pd.Series(clusters).map(lambda c: f"cluster {c}"))
    tab = pd.crosstab(pd.Series(names, name="cluster"),
                      pd.Series(pd.Categorical(labels, categories=ACTIVITY_ORDER), name="activity"),
                      dropna=False).reindex(columns=ACTIVITY_ORDER, fill_value=0)
    real = [r for r in tab.index if r != "noise"]
    real.sort(key=lambda r: (ACTIVITY_ORDER.index(tab.loc[r].idxmax()), -tab.loc[r].max()))
    return tab.loc[real + (["noise"] if "noise" in tab.index else [])]


def plot_contingency(tab, title, name):
    """Heatmap of counts; colour = share of each cluster's windows (row %), so purity is visible."""
    row_pct = tab.div(tab.sum(axis=1), axis=0) * 100
    fig, ax = plt.subplots(figsize=(10, 0.55 * len(tab) + 2.2))
    sns.heatmap(row_pct, annot=tab.values, fmt="d", cmap="Blues", vmin=0, vmax=100, linewidths=0.5,
                cbar_kws={"label": "% of cluster's windows"}, ax=ax)
    ax.set_title(title)
    ax.set_xlabel("True activity")
    ax.set_ylabel("Cluster (size)")
    ax.set_yticklabels([f"{r} ({n})" for r, n in zip(tab.index, tab.sum(axis=1))], rotation=0)
```

```
    plt.setp(ax.get_xticklabels(), rotation=30, ha="right", rotation_mode="anchor")
    save_fig(fig, name)
    plt.show()


def plot_clusters_pc12(clusters, title, name, noise_label=-1):
    """Train data on PC1/PC2 coloured by cluster; noise (if any) in grey underneath."""
    ids = [c for c in np.unique(clusters) if c != noise_label]
    colors = dict(zip(ids, sns.color_palette("husl", len(ids))))
    fig, ax = plt.subplots(figsize=(10, 7.5))
    m = clusters == noise_label
    if m.any():
        ax.scatter(Z2_tr[m, 0], Z2_tr[m, 1], s=6, color="lightgrey", alpha=0.7, edgecolors="none",
                   label=f"noise ({m.sum()})")
    for c in ids:
        m = clusters == c
        ax.scatter(Z2_tr[m, 0], Z2_tr[m, 1], s=6, alpha=0.6, color=colors[c], edgecolors="none",
                   label=f"cluster {c} ({m.sum()})")
    ax.set_title(title)
    ax.set_xlabel(f"PC1 ({100 * evr[0]:.1f}% of variance)")
    ax.set_ylabel(f"PC2 ({100 * evr[1]:.1f}% of variance)")
    ax.legend(title="Cluster (size)", markerscale=3, framealpha=0.9, fontsize=8)
    save_fig(fig, name)
    plt.show()
```

## 3.1 K-Means on all 561 standardized features: choosing k by silhouette

In [17]:

```
K_RANGE = range(2, 11)
kmeans_labels, rows = {}, []
for k in K_RANGE:
    t0 = time.perf_counter()
    km = KMeans(n_clusters=k, n_init=10, random_state=RANDOM_STATE).fit(X_tr)
    record_timing("phase3", f"kmeans_k{k}", time.perf_counter() - t0)
    kmeans_labels[k] = km.labels_
    rows.append({"k": k, "silhouette": silhouette_score(X_tr, km.labels_, **SIL_KW),
                 # float32 inertia wobbles in the 7th significant digit between runs; round it away
                 "inertia": round(float(km.inertia_)),
                 "ari_vs_activity": adjusted_rand_score(y_tr_arr, km.labels_)})
kmeans_sweep = pd.DataFrame(rows)
k_best = int(kmeans_sweep.loc[kmeans_sweep["silhouette"].idxmax(), "k"])
```

```
save_table(kmeans_sweep, "phase3_kmeans_silhouette", index=False)
display(kmeans_sweep.round(4))
print(f"Best k by silhouette: k = {k_best} (silhouette = {kmeans_sweep['silhouette'].max():.4f})")
print("(ARI is shown for reference only; k is chosen by silhouette alone.)")
```

```
saved results\phase3_kmeans_silhouette.csv
```

| | k | silhouette | inertia | ari_vs_activity |
|---|---|---|---|---|
| **0** | 2 | 0.4065 | 1741156 | 0.3292 |
| **1** | 3 | 0.3502 | 1556465 | 0.3252 |
| **2** | 4 | 0.1568 | 1481624 | 0.2904 |
| **3** | 5 | 0.1192 | 1409342 | 0.2943 |
| **4** | 6 | 0.1056 | 1364658 | 0.4310 |
| **5** | 7 | 0.1084 | 1332076 | 0.3525 |
| **6** | 8 | 0.1104 | 1296042 | 0.4250 |
| **7** | 9 | 0.0807 | 1271783 | 0.4332 |
| **8** | 10 | 0.1051 | 1244471 | 0.3736 |

```
Best k by silhouette: k = 2 (silhouette = 0.4065)
(ARI is shown for reference only; k is chosen by silhouette alone.)
```

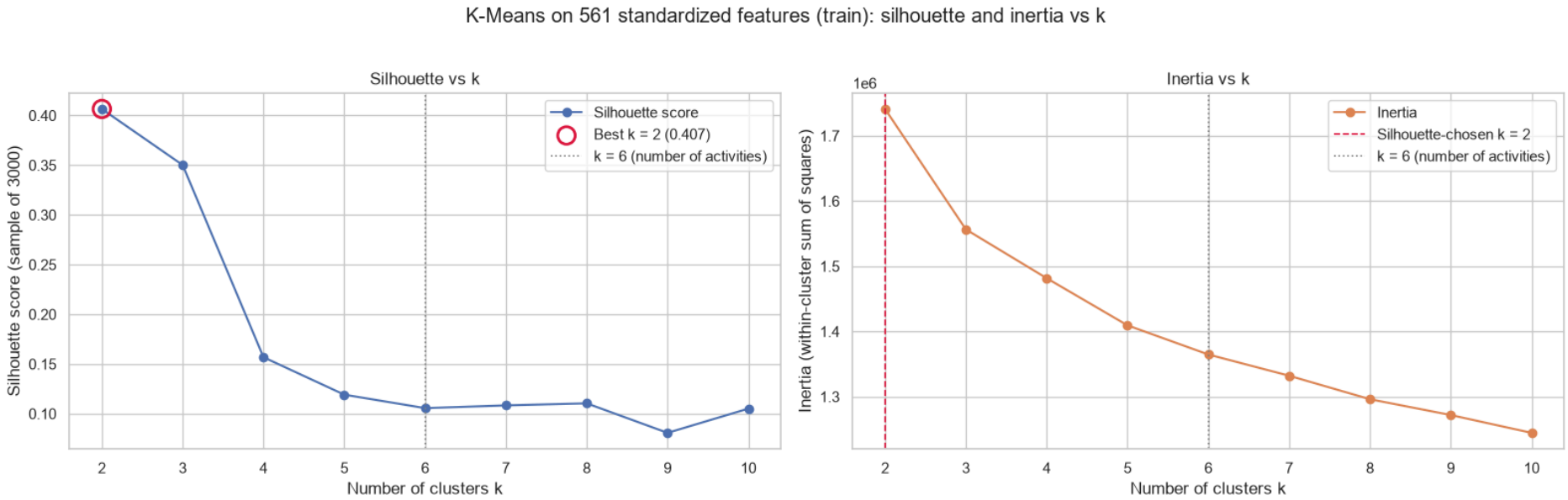
In [18]:

```
fig, (ax, ax2) = plt.subplots(1, 2, figsize=(16, 5))
ax.plot(kmeans_sweep["k"], kmeans_sweep["silhouette"], marker="o", color="#4C72B0", label="Silhouette score")
best = kmeans_sweep.set_index("k").loc[k_best, "silhouette"]
ax.scatter([k_best], [best], s=160, facecolor="none", edgecolor="crimson", lw=2, zorder=5,
           label=f"Best k = {k_best} ({best:.3f})")
ax.axvline(6, color="gray", ls=":", lw=1.2, label="k = 6 (number of activities)")
ax.set_title("Silhouette vs k")
ax.set_xlabel("Number of clusters k")
ax.set_ylabel("Silhouette score (sample of 3000)")
ax.set_xticks(list(K_RANGE))
ax.legend()

ax2.plot(kmeans_sweep["k"], kmeans_sweep["inertia"], marker="o", color="#DD8452", label="Inertia")
ax2.axvline(k_best, color="crimson", ls="--", lw=1.2, label=f"Silhouette-chosen k = {k_best}")
ax2.axvline(6, color="gray", ls=":", lw=1.2, label="k = 6 (number of activities)")
ax2.set_title("Inertia vs k")
ax2.set_xlabel("Number of clusters k")
ax2.set_ylabel("Inertia (within-cluster sum of squares)")
ax2.set_xticks(list(K_RANGE))
ax2.legend()
fig.suptitle("K-Means on 561 standardized features (train): silhouette and inertia vs k", y=1.02)
fig.tight_layout()
```

```
save_fig(fig, "phase3_kmeans_silhouette_inertia_vs_k")
plt.show()
```

```
saved figures\phase3_kmeans_silhouette_inertia_vs_k.png
```

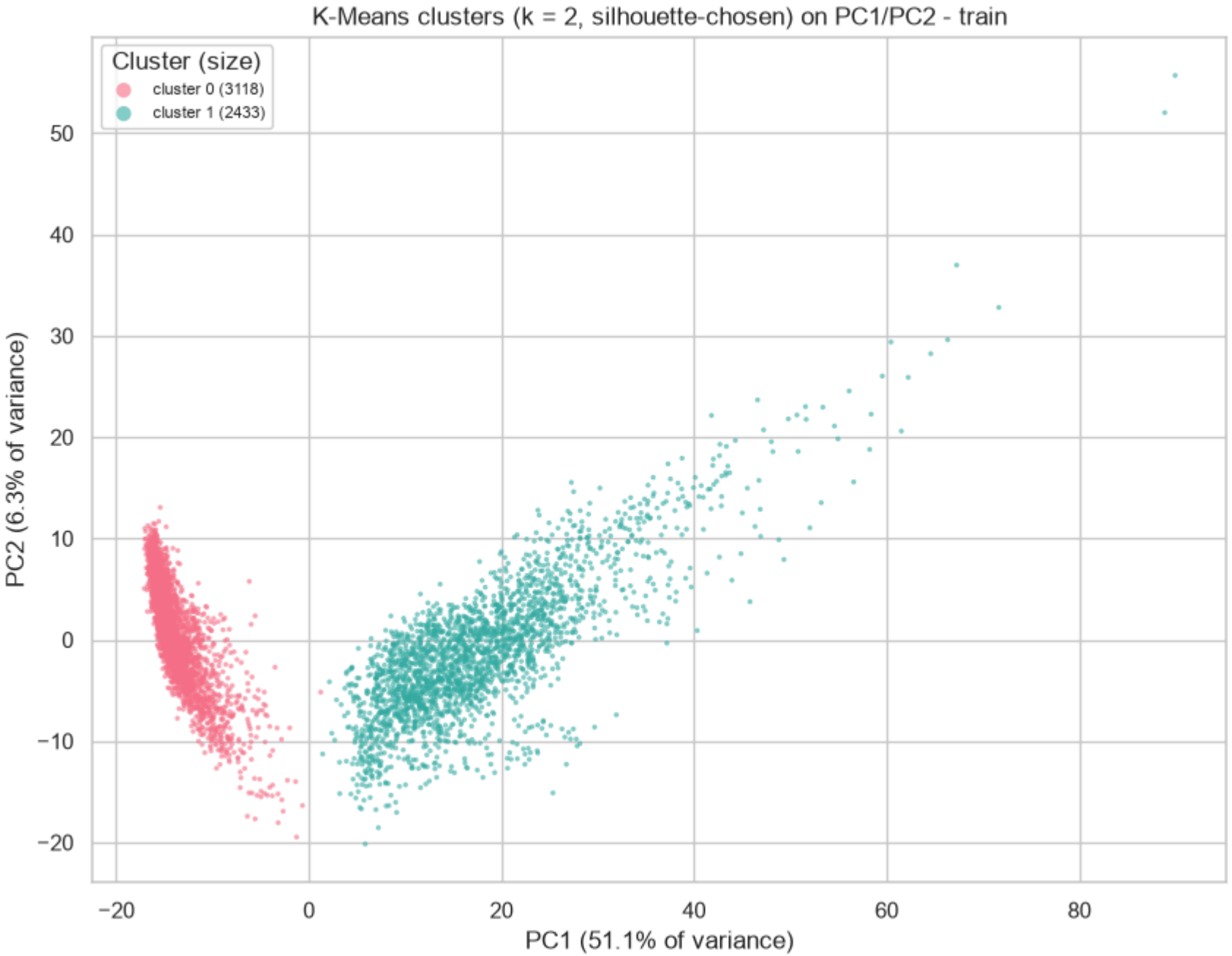



## 3.2 K-Means with the silhouette-chosen k

In [19]:
```
lab_best = kmeans_labels[k_best]
ari_best = adjusted_rand_score(y_tr_arr, lab_best)
print(f"K-Means k = {k_best}: ARI vs true activity = {ari_best:.4f}")

plot_clusters_pc12(lab_best, f"K-Means clusters (k = {k_best}, silhouette-chosen) on PC1/PC2 - train",
                   "phase3_kmeans_best_pc12")
tab_best = contingency(lab_best, y_tr_arr)
save_table(tab_best, f"phase3_kmeans_k{k_best}_contingency")
plot_contingency(tab_best, f"K-Means k = {k_best} (silhouette-chosen) vs true activity - ARI = {ari_best:.3f}",
                 f"phase3_kmeans_k{k_best}_contingency")
```

```
K-Means k = 2: ARI vs true activity = 0.3292
saved figures\phase3_kmeans_best_pc12.png
```




saved results\phase3_kmeans_k2_contingency.csv
saved figures\phase3_kmeans_k2_contingency.png




## 3.3 K-Means with k = 6 (the true number of activities)

In [20]:
```
lab_k6 = kmeans_labels[6]
ari_k6 = adjusted_rand_score(y_tr_arr, lab_k6)
print(f"K-Means k = 6: ARI vs true activity = {ari_k6:.4f}")

tab_k6 = contingency(lab_k6, y_tr_arr)
save_table(tab_k6, "phase3_kmeans_k6_contingency")
plot_contingency(tab_k6, f"K-Means k = 6 vs true activity - ARI = {ari_k6:.3f}", "phase3_kmeans_k6_contingency")

kmeans_compare = kmeans_sweep.set_index("k").loc[[k_best, 6], ["silhouette", "ari_vs_activity"]]
kmeans_compare.insert(0, "choice", ["silhouette-chosen" if k_best != 6 else "silhouette-chosen = natural",
                                    "natural (6 activities)"][:len(kmeans_compare)])
```

```
save_table(kmeans_compare, "phase3_kmeans_best_vs_k6")
display(kmeans_compare.round(4))
```

```
K-Means k = 6: ARI vs true activity = 0.4310
saved results\phase3_kmeans_k6_contingency.csv
saved figures\phase3_kmeans_k6_contingency.png
```




```
saved results\phase3_kmeans_best_vs_k6.csv
```

| k | choice | silhouette | ari_vs_activity |
|---|---|---|---|
| **2** | silhouette-chosen | 0.4065 | 0.3292 |
| **6** | natural (6 activities) | 0.1056 | 0.4310 |

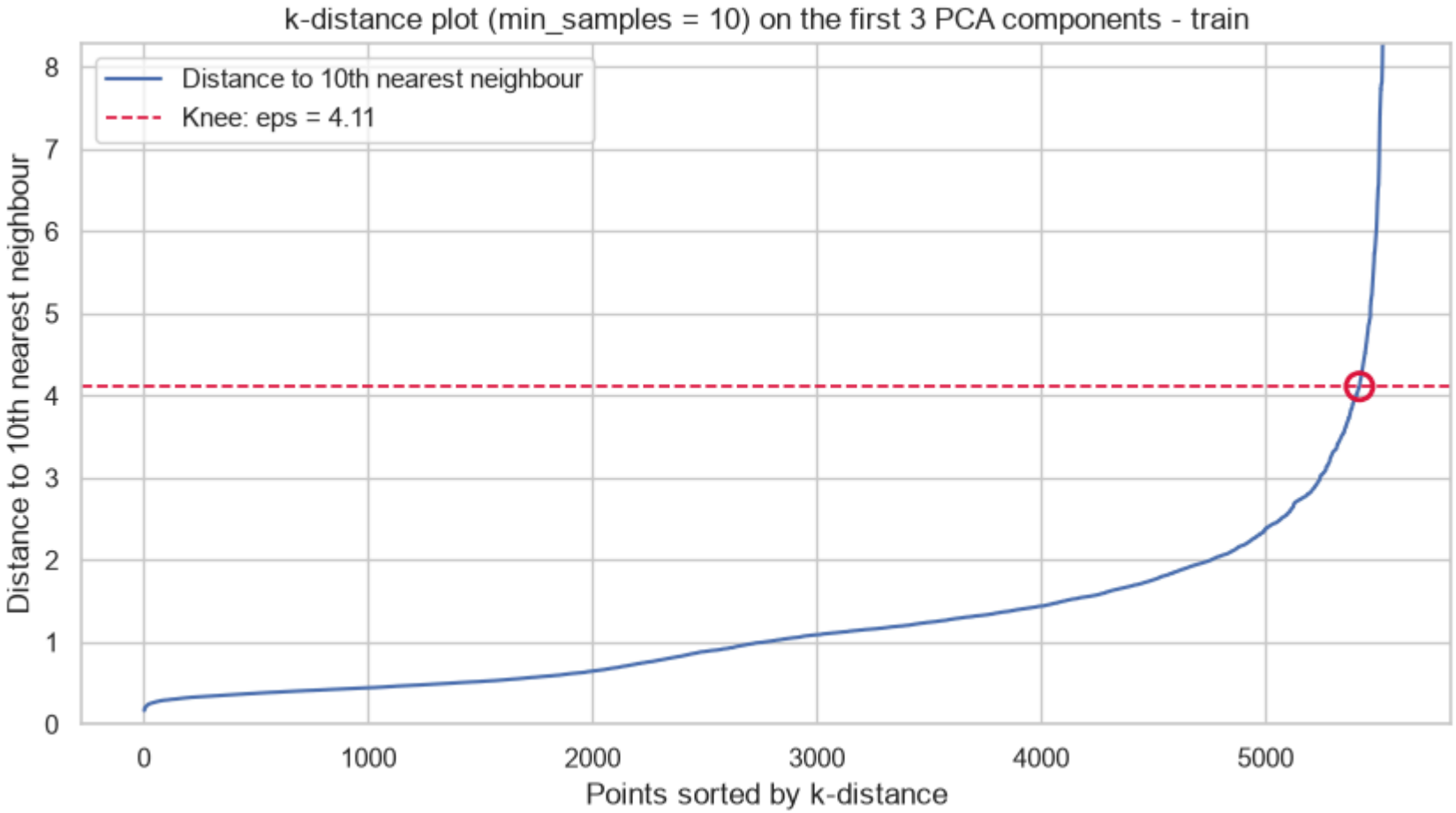
## 3.4 DBSCAN on the top `k_elbow` PCA components: k-distance plot

In [21]:
```
MIN_SAMPLES = 10
Zk_tr = pca.transform(X_tr)[:, :k_elbow]
print(f"DBSCAN space: first {k_elbow} PCA components -> {Zk_tr.shape}")

# Distance to the min_samples-th nearest neighbour. kneighbors on the fitted data returns the point itself
# as its first neighbour, which matches DBSCAN's min_samples (that count includes the point itself).
nn = NearestNeighbors(n_neighbors=MIN_SAMPLES).fit(Zk_tr)
kdist = np.sort(nn.kneighbors(Zk_tr)[0][:, -1])
knee_idx, _ = elbow_max_distance(kdist)          # same max-distance-to-line method as Phase 2
eps_knee = float(kdist[knee_idx - 1])
print(f"k-distance knee: eps = {eps_knee:.3f} (at sorted point {knee_idx} of {len(kdist)})")

fig, ax = plt.subplots(figsize=(10, 5))
ax.plot(np.arange(1, len(kdist) + 1), kdist, color="#4C72B0", lw=1.5,
        label=f"Distance to {MIN_SAMPLES}th nearest neighbour")
ax.axhline(eps_knee, color="crimson", ls="--", lw=1.2, label=f"Knee: eps = {eps_knee:.2f}")
ax.scatter([knee_idx], [eps_knee], s=120, facecolor="none", edgecolor="crimson", lw=2, zorder=5)
ax.set_title(f"k-distance plot (min_samples = {MIN_SAMPLES}) on the first {k_elbow} PCA components - train")
ax.set_xlabel("Points sorted by k-distance")
ax.set_ylabel(f"Distance to {MIN_SAMPLES}th nearest neighbour")
ax.set_ylim(0, np.percentile(kdist, 99.5))   # cap at the 99.5th percentile so outliers don't squash the knee
ax.legend()
save_fig(fig, "phase3_dbscan_k_distance")
plt.show()
```

```
DBSCAN space: first 3 PCA components -> (5551, 3)
k-distance knee: eps = 4.111 (at sorted point 5419 of 5551)
saved figures\phase3_dbscan_k_distance.png
```




## 3.5 DBSCAN eps sweep around the knee

Silhouette is computed in the same PCA space DBSCAN clusters in, on non-noise points only, and only when there are at least 2 clusters. ARI treats noise as one extra group (all points); `ari_non_noise` is also reported.

In [22]:
```
eps_grid = np.round(np.linspace(0.5 * eps_knee, 2.0 * eps_knee, 31), 4)
rows, dbscan_labels = [], {}
for eps in eps_grid:
    t0 = time.perf_counter()
    lab = DBSCAN(eps=eps, min_samples=MIN_SAMPLES).fit_predict(Zk_tr)
    record_timing("phase3", f"dbscan_eps{eps}", time.perf_counter() - t0)
```

```
    dbscan_labels[eps] = lab
    core = lab != -1
    n_clusters = len(set(lab[core]))
    sil = np.nan
    if n_clusters >= 2:
        sil = silhouette_score(Zk_tr[core], lab[core], **SIL_KW)
    rows.append({"eps": eps, "n_clusters": n_clusters, "noise_pct": 100 * (~core).mean(), "silhouette": sil,
                 "ari": adjusted_rand_score(y_tr_arr, lab),
                 "ari_non_noise": adjusted_rand_score(y_tr_arr[core], lab[core]) if core.any() else np.nan})
dbscan_sweep = pd.DataFrame(rows)

MAX_NOISE_PCT = 10.0
eligible = dbscan_sweep[(dbscan_sweep["n_clusters"] >= 2) & (dbscan_sweep["noise_pct"] <= MAX_NOISE_PCT)
                        & dbscan_sweep["silhouette"].notna()]
assert len(eligible), "no eps gives >= 2 clusters with <= 10% noise; widen the eps grid"
best_row = eligible.loc[eligible["silhouette"].idxmax()]
eps_best = float(best_row["eps"])
dbscan_sweep["selected"] = dbscan_sweep["eps"] == eps_best

save_table(dbscan_sweep, "phase3_dbscan_eps_sweep", index=False)
display(dbscan_sweep.round(4))
print(f"Best eps (max silhouette, >= 2 clusters, noise <= {MAX_NOISE_PCT:.0f}%): eps = {eps_best:.3f} -> "
      f"{int(best_row['n_clusters'])} clusters, {best_row['noise_pct']:.1f}% noise, "
      f"silhouette {best_row['silhouette']:.4f}, ARI {best_row['ari']:.4f}")
```

```
saved results\phase3_dbscan_eps_sweep.csv
```

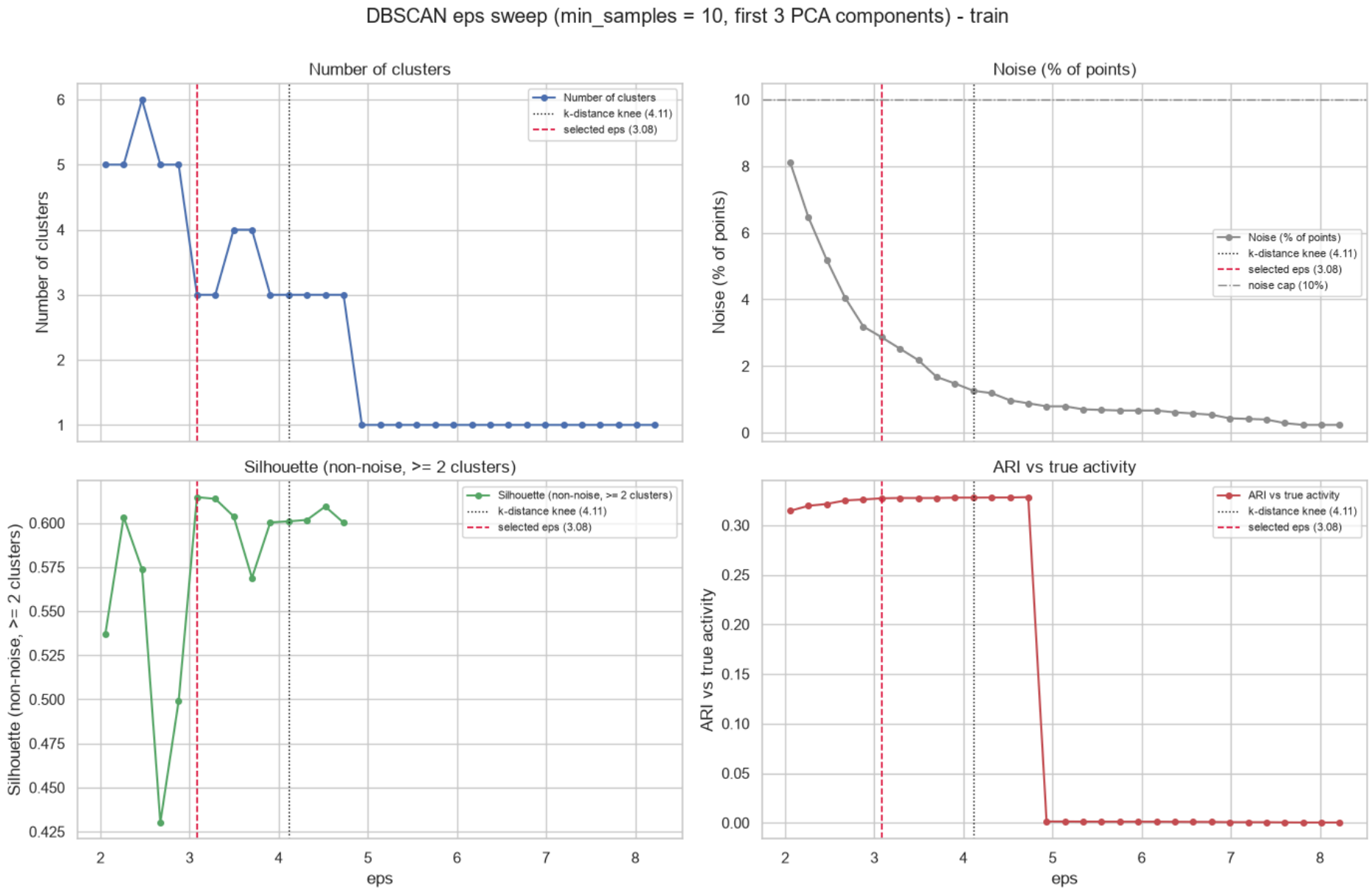
| | eps | n_clusters | noise_pct | silhouette | ari | ari_non_noise | selected |
|---|---|---|---|---|---|---|---|
| **0** | 2.0554 | 5 | 8.1247 | 0.5373 | 0.3145 | 0.3240 | False |
| **1** | 2.2609 | 5 | 6.4673 | 0.6034 | 0.3195 | 0.3265 | False |
| **2** | 2.4665 | 6 | 5.1883 | 0.5738 | 0.3214 | 0.3268 | False |
| **3** | 2.6720 | 5 | 4.0533 | 0.4302 | 0.3249 | 0.3280 | False |
| **4** | 2.8775 | 5 | 3.1886 | 0.4991 | 0.3259 | 0.3282 | False |
| **5** | 3.0831 | 3 | 2.8643 | 0.6146 | 0.3271 | 0.3290 | True |
| **6** | 3.2886 | 3 | 2.5221 | 0.6137 | 0.3274 | 0.3290 | False |
| **7** | 3.4942 | 4 | 2.1798 | 0.6038 | 0.3274 | 0.3289 | False |
| **8** | 3.6997 | 4 | 1.6754 | 0.5689 | 0.3274 | 0.3290 | False |
| **9** | 3.9052 | 3 | 1.4772 | 0.6003 | 0.3278 | 0.3291 | False |
| **10** | 4.1108 | 3 | 1.2610 | 0.6009 | 0.3278 | 0.3289 | False |
| **11** | 4.3163 | 3 | 1.1890 | 0.6017 | 0.3279 | 0.3289 | False |
| **12** | 4.5219 | 3 | 0.9728 | 0.6093 | 0.3279 | 0.3287 | False |
| **13** | 4.7274 | 3 | 0.8827 | 0.6002 | 0.3283 | 0.3290 | False |
| **14** | 4.9329 | 1 | 0.7926 | NaN | 0.0013 | 0.0000 | False |
| **15** | 5.1385 | 1 | 0.7926 | NaN | 0.0013 | 0.0000 | False |
| **16** | 5.3440 | 1 | 0.7026 | NaN | 0.0012 | 0.0000 | False |
| **17** | 5.5496 | 1 | 0.6846 | NaN | 0.0012 | 0.0000 | False |
| **18** | 5.7551 | 1 | 0.6665 | NaN | 0.0011 | 0.0000 | False |
| **19** | 5.9606 | 1 | 0.6665 | NaN | 0.0011 | 0.0000 | False |
| **20** | 6.1662 | 1 | 0.6665 | NaN | 0.0011 | 0.0000 | False |
| **21** | 6.3717 | 1 | 0.6125 | NaN | 0.0010 | 0.0000 | False |
| **22** | 6.5772 | 1 | 0.5765 | NaN | 0.0009 | 0.0000 | False |
| **23** | 6.7828 | 1 | 0.5404 | NaN | 0.0009 | 0.0000 | False |
| **24** | 6.9883 | 1 | 0.4324 | NaN | 0.0006 | 0.0000 | False |
| **25** | 7.1939 | 1 | 0.4143 | NaN | 0.0006 | 0.0000 | False |
| **26** | 7.3994 | 1 | 0.3963 | NaN | 0.0006 | 0.0000 | False |
| **27** | 7.6049 | 1 | 0.2882 | NaN | 0.0004 | 0.0000 | False |
| **28** | 7.8105 | 1 | 0.2342 | NaN | 0.0003 | 0.0000 | False |
| **29** | 8.0160 | 1 | 0.2342 | NaN | 0.0003 | 0.0000 | False |
| **30** | 8.2216 | 1 | 0.2342 | NaN | 0.0003 | 0.0000 | False |

```
Best eps (max silhouette, >= 2 clusters, noise <= 10%): eps = 3.083 -> 3 clusters, 2.9% noise, silhouette 0.6146, ARI 0.3271
```

In [23]:
```python
fig, axes = plt.subplots(2, 2, figsize=(14, 9), sharex=True)
panels = [("n_clusters", "Number of clusters", "#4C72B0"), ("noise_pct", "Noise (% of points)", "#8C8C8C"),
          ("silhouette", "Silhouette (non-noise, >= 2 clusters)", "#55A868"), ("ari", "ARI vs true activity", "#C44E52")]
for ax, (col, label, color) in zip(axes.ravel(), panels):
    ax.plot(dbscan_sweep["eps"], dbscan_sweep[col], marker="o", ms=4, color=color, label=label)
    ax.axvline(eps_knee, color="black", ls=":", lw=1, label=f"k-distance knee ({eps_knee:.2f})")
    ax.axvline(eps_best, color="crimson", ls="--", lw=1.2, label=f"selected eps ({eps_best:.2f})")
    if col == "noise_pct":
        ax.axhline(MAX_NOISE_PCT, color="gray", ls="-.", lw=1, label=f"noise cap ({MAX_NOISE_PCT:.0f}%)")
```

```
    ax.set_title(label)
    ax.set_ylabel(label)
    ax.legend(fontsize=8)
for ax in axes[1]:
    ax.set_xlabel("eps")
fig.suptitle(f"DBSCAN eps sweep (min_samples = {MIN_SAMPLES}, first {k_elbow} PCA components) - train", y=1.0)
fig.tight_layout()
save_fig(fig, "phase3_dbscan_eps_sweep")
plt.show()
```

```
saved figures\phase3_dbscan_eps_sweep.png
```

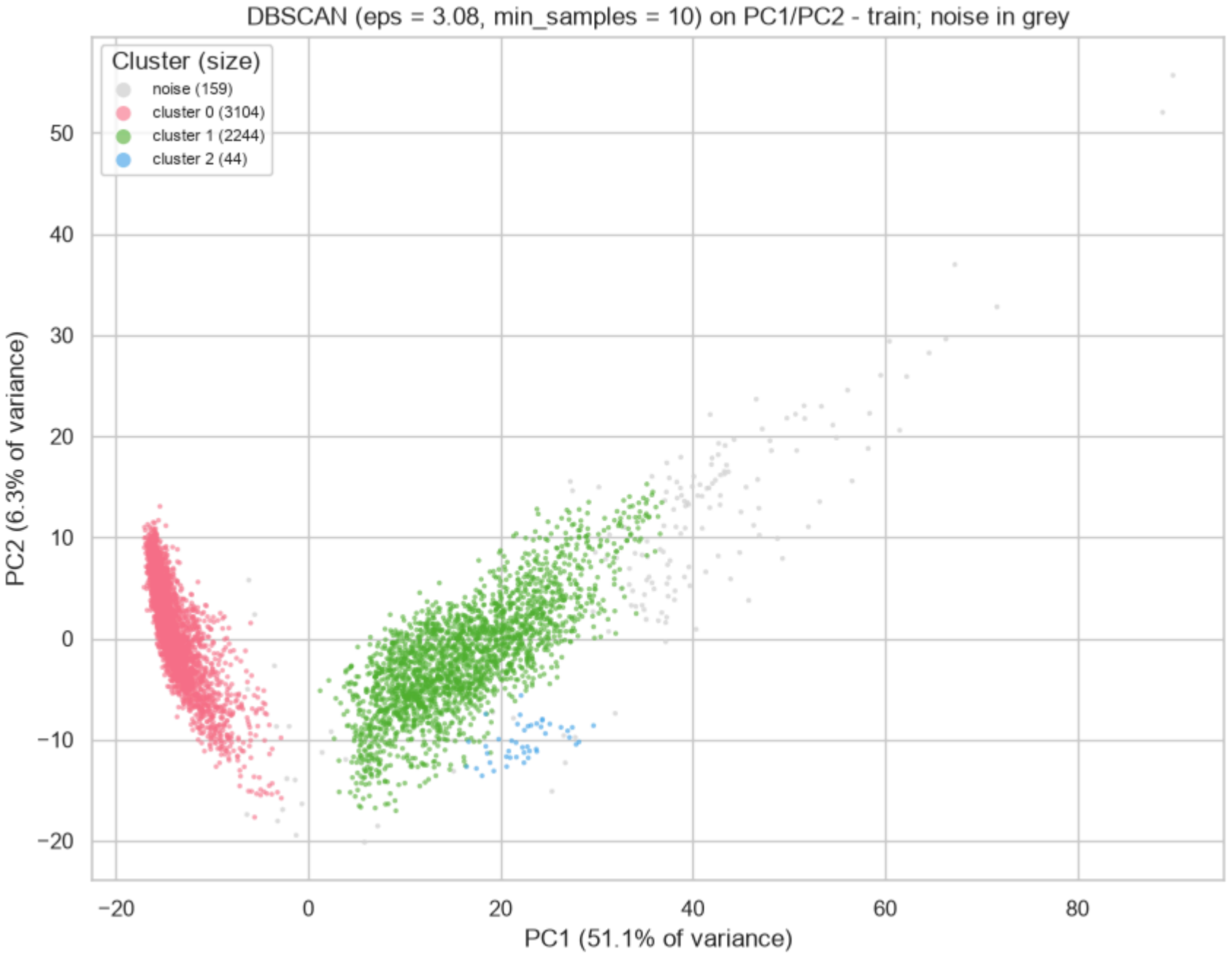



## 3.6 Best DBSCAN setting

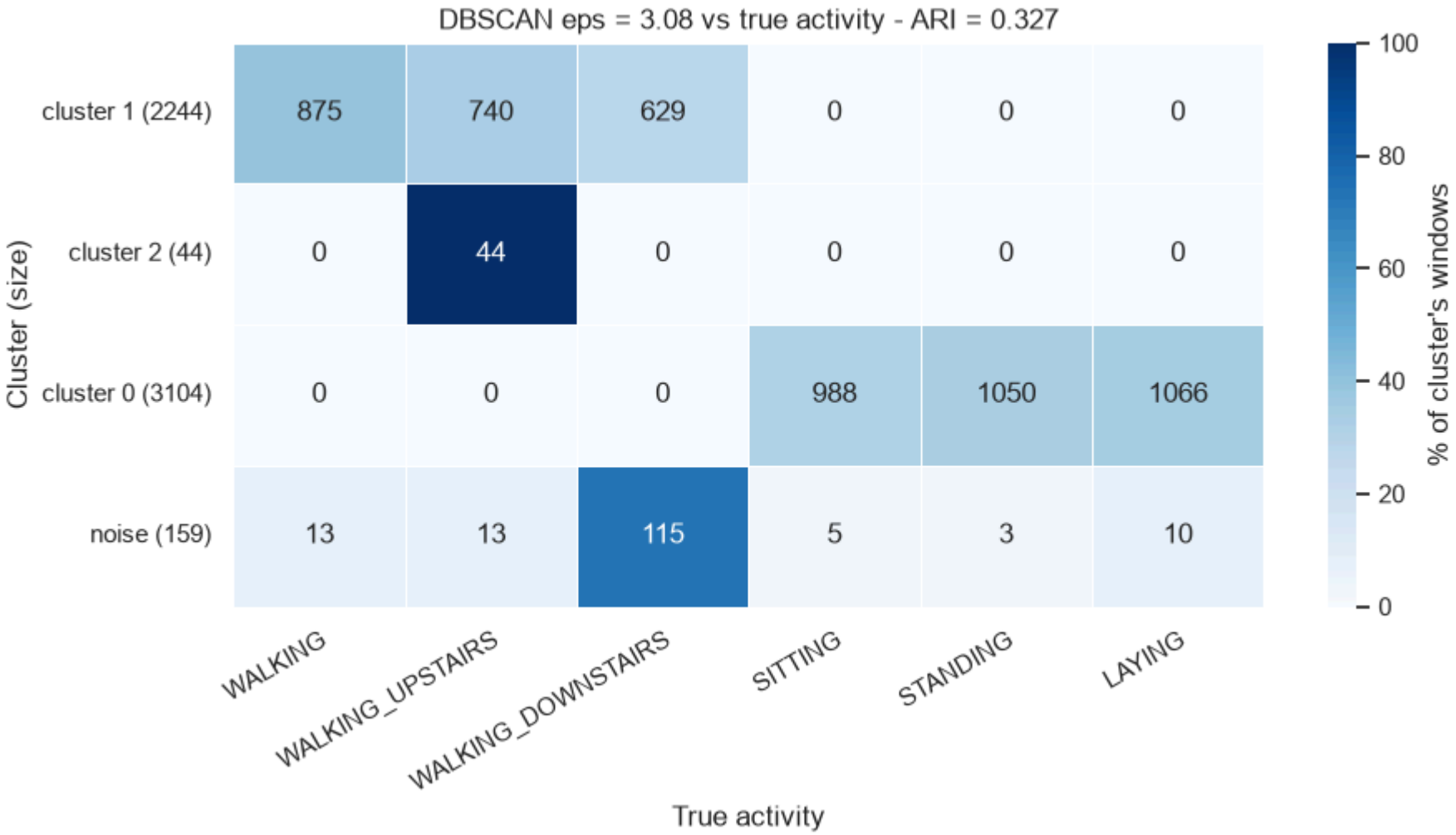
In [24]:
```
lab_db = dbscan_labels[eps_best]
ari_db = adjusted_rand_score(y_tr_arr, lab_db)
```

```
plot_clusters_pc12(lab_db, f"DBSCAN (eps = {eps_best:.2f}, min_samples = {MIN_SAMPLES}) on PC1/PC2 - train; noise in grey",
                   "phase3_dbscan_best_pc12")
tab_db = contingency(lab_db, y_tr_arr)
save_table(tab_db, "phase3_dbscan_best_contingency")
plot_contingency(tab_db, f"DBSCAN eps = {eps_best:.2f} vs true activity - ARI = {ari_db:.3f}",
                 "phase3_dbscan_best_contingency")
```

```
saved figures\phase3_dbscan_best_pc12.png
```




```
saved results\phase3_dbscan_best_contingency.csv
saved figures\phase3_dbscan_best_contingency.png
```




In [25]:
```
# Summary of the three clusterings compared in this phase
phase3_summary = pd.DataFrame([
    {"method": f"K-Means k={k_best} (silhouette-chosen)", "space": "561 standardized features",
     "n_clusters": k_best, "noise_pct": 0.0,
     "silhouette": kmeans_sweep.set_index("k").loc[k_best, "silhouette"], "ari": ari_best},
    {"method": "K-Means k=6 (natural)", "space": "561 standardized features", "n_clusters": 6, "noise_pct": 0.0,
     "silhouette": kmeans_sweep.set_index("k").loc[6, "silhouette"], "ari": ari_k6},
    {"method": f"DBSCAN eps={eps_best:.2f}, min_samples={MIN_SAMPLES}", "space": f"first {k_elbow} PCA components",
     "n_clusters": int(best_row["n_clusters"]), "noise_pct": best_row["noise_pct"],
     "silhouette": best_row["silhouette"], "ari": ari_db},
```

```
])
save_table(phase3_summary, "phase3_summary", index=False)
display(phase3_summary.round(4))
print("Note: silhouettes are computed in each method's own space, so compare them within a method, not across.")
```

```
saved results\phase3_summary.csv
```

| | method | space | n_clusters | noise_pct | silhouette | ari |
|---|---|---|---|---|---|---|
| **0** | K-Means k=2 (silhouette-chosen) | 561 standardized features | 2 | 0.0000 | 0.4065 | 0.3292 |
| **1** | K-Means k=6 (natural) | 561 standardized features | 6 | 0.0000 | 0.1056 | 0.4310 |
| **2** | DBSCAN eps=3.08, min_samples=10 | first 3 PCA components | 3 | 2.8643 | 0.6146 | 0.3271 |

```
Note: silhouettes are computed in each method's own space, so compare them within a method, not across.
```

## 3.7 Comparison: DBSCAN on all 561 standardized features

The same procedure as 3.4-3.6 ( `min_samples=10` , k-distance knee, eps swept from 0.5x to 2x the knee, same selection rule), but on all 561 standardized features instead of the first `k_elbow` PCA components. If no eps gives >= 2 clusters with <= 10% noise, this is stated and the best available setting is reported instead: the highest silhouette among settings with >= 2 clusters (noise cap relaxed), or, if every eps gives at most one cluster, the lowest-noise setting. **The 3-PC run above remains the main DBSCAN result**; this section is only a comparison.

In [26]:
```
def k_distance_knee(Z, min_samples=MIN_SAMPLES):
    """Sorted distance to the min_samples-th neighbour (self included, as in DBSCAN) and its knee."""
    kd = np.sort(NearestNeighbors(n_neighbors=min_samples).fit(Z).kneighbors(Z)[0][:, -1])
    idx, _ = elbow_max_distance(kd)
    return kd, idx, float(kd[idx - 1])


def run_dbscan_sweep(Z, eps_grid, timing_tag, min_samples=MIN_SAMPLES):
    """DBSCAN for each eps; silhouette on non-noise points in Z (only with >= 2 clusters); ARI with noise as a group."""
    rows, labels = [], {}
    for eps in eps_grid:
        t0 = time.perf_counter()
        lab = DBSCAN(eps=eps, min_samples=min_samples).fit_predict(Z)
        record_timing("phase3", f"{timing_tag}_eps{eps}", time.perf_counter() - t0)
        labels[eps] = lab
        core = lab != -1
        n_clusters = len(set(lab[core]))
        rows.append({"eps": eps, "n_clusters": n_clusters, "noise_pct": 100 * (~core).mean(),
                     "silhouette": silhouette_score(Z[core], lab[core], **SIL_KW) if n_clusters >= 2 else np.nan,
                     "ari": adjusted_rand_score(y_tr_arr, lab),
                     "ari_non_noise": adjusted_rand_score(y_tr_arr[core], lab[core]) if core.any() else np.nan})
```

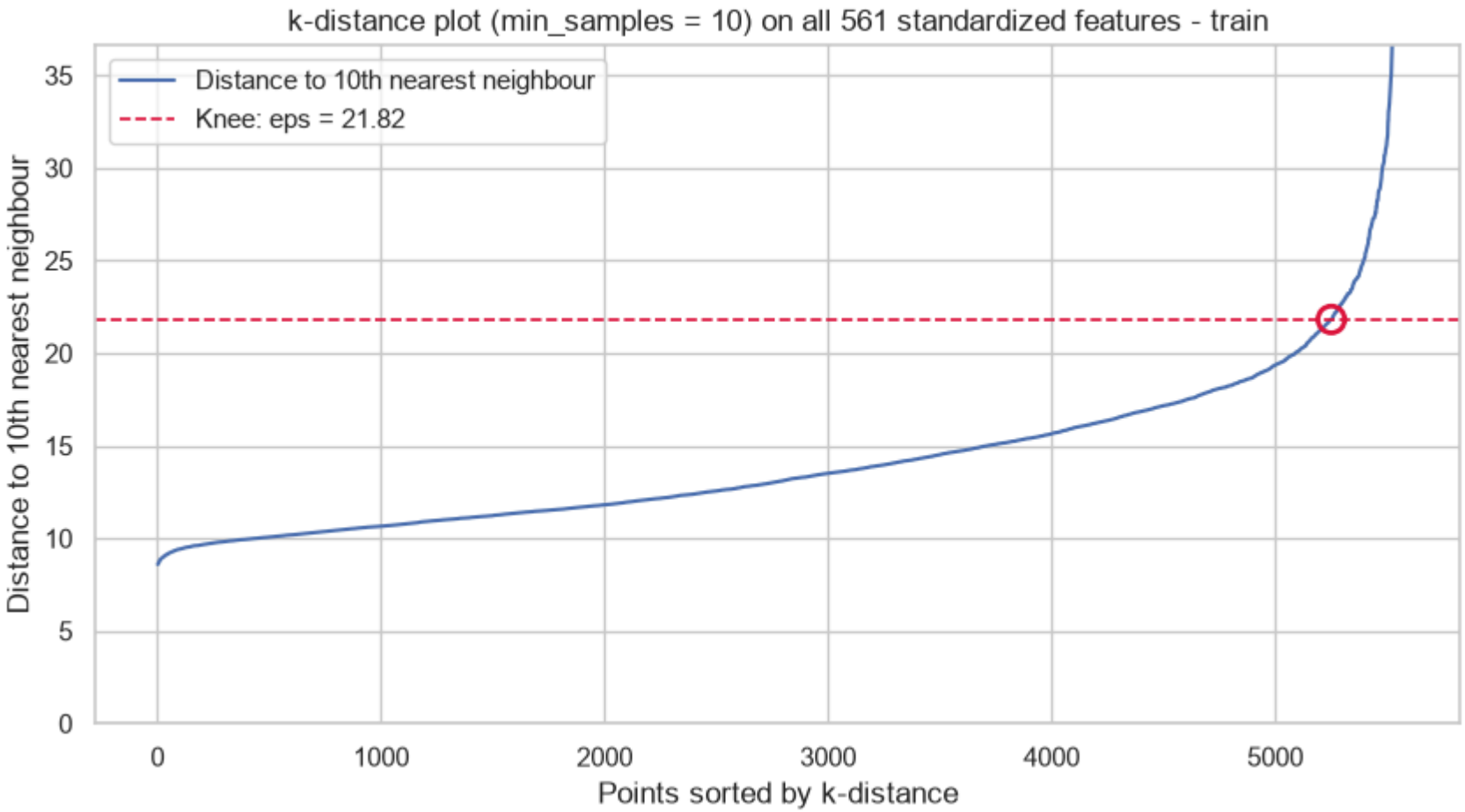
```
    return pd.DataFrame(rows), labels


def select_eps(sweep, max_noise_pct=MAX_NOISE_PCT):
    """Same rule as the 3-PC run, with an explicit fallback. Returns (row, rule_satisfied, explanation)."""
    multi = sweep[(sweep["n_clusters"] >= 2) & sweep["silhouette"].notna()]
    ok = multi[multi["noise_pct"] <= max_noise_pct]
    if len(ok):
        return ok.loc[ok["silhouette"].idxmax()], True, f"max silhouette with >= 2 clusters and noise <= {max_noise_pct:.0f}%"
    if len(multi):
        return (multi.loc[multi["silhouette"].idxmax()], False,
                f"NO eps qualifies (>= 2 clusters with noise <= {max_noise_pct:.0f}%); best available = "
                "max silhouette among >= 2-cluster settings, noise cap relaxed")
    return (sweep.loc[sweep["noise_pct"].idxmin()], False,
            "NO eps gives >= 2 clusters; best available = lowest-noise setting (a single cluster)")


kdist561, knee_idx561, eps_knee561 = k_distance_knee(X_tr)
print(f"561-D k-distance knee: eps = {eps_knee561:.3f} (at sorted point {knee_idx561} of {len(kdist561)})")

fig, ax = plt.subplots(figsize=(10, 5))
ax.plot(np.arange(1, len(kdist561) + 1), kdist561, color="#4C72B0", lw=1.5,
        label=f"Distance to {MIN_SAMPLES}th nearest neighbour")
ax.axhline(eps_knee561, color="crimson", ls="--", lw=1.2, label=f"Knee: eps = {eps_knee561:.2f}")
ax.scatter([knee_idx561], [eps_knee561], s=120, facecolor="none", edgecolor="crimson", lw=2, zorder=5)
ax.set_title(f"k-distance plot (min_samples = {MIN_SAMPLES}) on all 561 standardized features - train")
ax.set_xlabel("Points sorted by k-distance")
ax.set_ylabel(f"Distance to {MIN_SAMPLES}th nearest neighbour")
ax.set_ylim(0, np.percentile(kdist561, 99.5))   # same 99.5th-percentile cap as the 3-PC plot
ax.legend()
save_fig(fig, "phase3_dbscan561_k_distance")
plt.show()
```

```
561-D k-distance knee: eps = 21.819 (at sorted point 5251 of 5551)
saved figures\phase3_dbscan561_k_distance.png
```




In [27]:
```
eps_grid561 = np.round(np.linspace(0.5 * eps_knee561, 2.0 * eps_knee561, 31), 4)
dbscan561_sweep, dbscan561_labels = run_dbscan_sweep(X_tr, eps_grid561, "dbscan561")
best561, rule_ok561, why561 = select_eps(dbscan561_sweep)
eps_best561 = float(best561["eps"])
dbscan561_sweep["selected"] = dbscan561_sweep["eps"] == eps_best561

save_table(dbscan561_sweep, "phase3_dbscan561_eps_sweep", index=False)
display(dbscan561_sweep.round(4))
print(f"Selection: {why561}")
print(f"-> eps = {eps_best561:.3f}: {int(best561['n_clusters'])} cluster(s), {best561['noise_pct']:.1f}% noise, "
      f"silhouette {best561['silhouette']:.4f}, ARI {best561['ari']:.4f}")
```

```
saved results\phase3_dbscan561_eps_sweep.csv
```

| | eps | n_clusters | noise_pct | silhouette | ari | ari_non_noise | selected |
|---|---|---|---|---|---|---|---|
| **0** | 10.9097 | 10 | 63.7182 | 0.0533 | 0.0957 | 0.0863 | False |
| **1** | 12.0006 | 11 | 49.0542 | 0.3671 | 0.1877 | 0.1759 | False |
| **2** | 13.0916 | 12 | 38.3534 | 0.3500 | 0.2484 | 0.2422 | False |
| **3** | 14.1826 | 4 | 29.4361 | 0.3326 | 0.3014 | 0.2986 | False |
| **4** | 15.2735 | 9 | 20.8971 | 0.3277 | 0.3117 | 0.3113 | False |
| **5** | 16.3645 | 4 | 14.7181 | 0.3792 | 0.3226 | 0.3216 | False |
| **6** | 17.4555 | 2 | 9.8181 | 0.3090 | 0.0266 | 0.0017 | True |
| **7** | 18.5465 | 1 | 7.1338 | NaN | 0.0161 | 0.0000 | False |
| **8** | 19.6374 | 1 | 5.3144 | NaN | 0.0107 | 0.0000 | False |
| **9** | 20.7284 | 1 | 4.0713 | NaN | 0.0077 | 0.0000 | False |
| **10** | 21.8194 | 1 | 3.0625 | NaN | 0.0054 | 0.0000 | False |
| **11** | 22.9103 | 1 | 2.5401 | NaN | 0.0044 | 0.0000 | False |
| **12** | 24.0013 | 1 | 2.1077 | NaN | 0.0036 | 0.0000 | False |
| **13** | 25.0923 | 1 | 1.5853 | NaN | 0.0026 | 0.0000 | False |
| **14** | 26.1832 | 1 | 1.2250 | NaN | 0.0020 | 0.0000 | False |
| **15** | 27.2742 | 1 | 0.9908 | NaN | 0.0016 | 0.0000 | False |
| **16** | 28.3652 | 1 | 0.8107 | NaN | 0.0012 | 0.0000 | False |
| **17** | 29.4561 | 1 | 0.7206 | NaN | 0.0010 | 0.0000 | False |
| **18** | 30.5471 | 1 | 0.5404 | NaN | 0.0007 | 0.0000 | False |
| **19** | 31.6381 | 1 | 0.4324 | NaN | 0.0006 | 0.0000 | False |
| **20** | 32.7290 | 1 | 0.3243 | NaN | 0.0005 | 0.0000 | False |
| **21** | 33.8200 | 1 | 0.2522 | NaN | 0.0003 | 0.0000 | False |
| **22** | 34.9110 | 1 | 0.1801 | NaN | 0.0002 | 0.0000 | False |
| **23** | 36.0019 | 1 | 0.1621 | NaN | 0.0002 | 0.0000 | False |
| **24** | 37.0929 | 1 | 0.1261 | NaN | 0.0002 | 0.0000 | False |
| **25** | 38.1839 | 1 | 0.0901 | NaN | 0.0001 | 0.0000 | False |
| **26** | 39.2748 | 1 | 0.0901 | NaN | 0.0001 | 0.0000 | False |
| **27** | 40.3658 | 1 | 0.0901 | NaN | 0.0001 | 0.0000 | False |
| **28** | 41.4568 | 1 | 0.0721 | NaN | 0.0001 | 0.0000 | False |
| **29** | 42.5477 | 1 | 0.0540 | NaN | 0.0001 | 0.0000 | False |
| **30** | 43.6387 | 1 | 0.0540 | NaN | 0.0001 | 0.0000 | False |

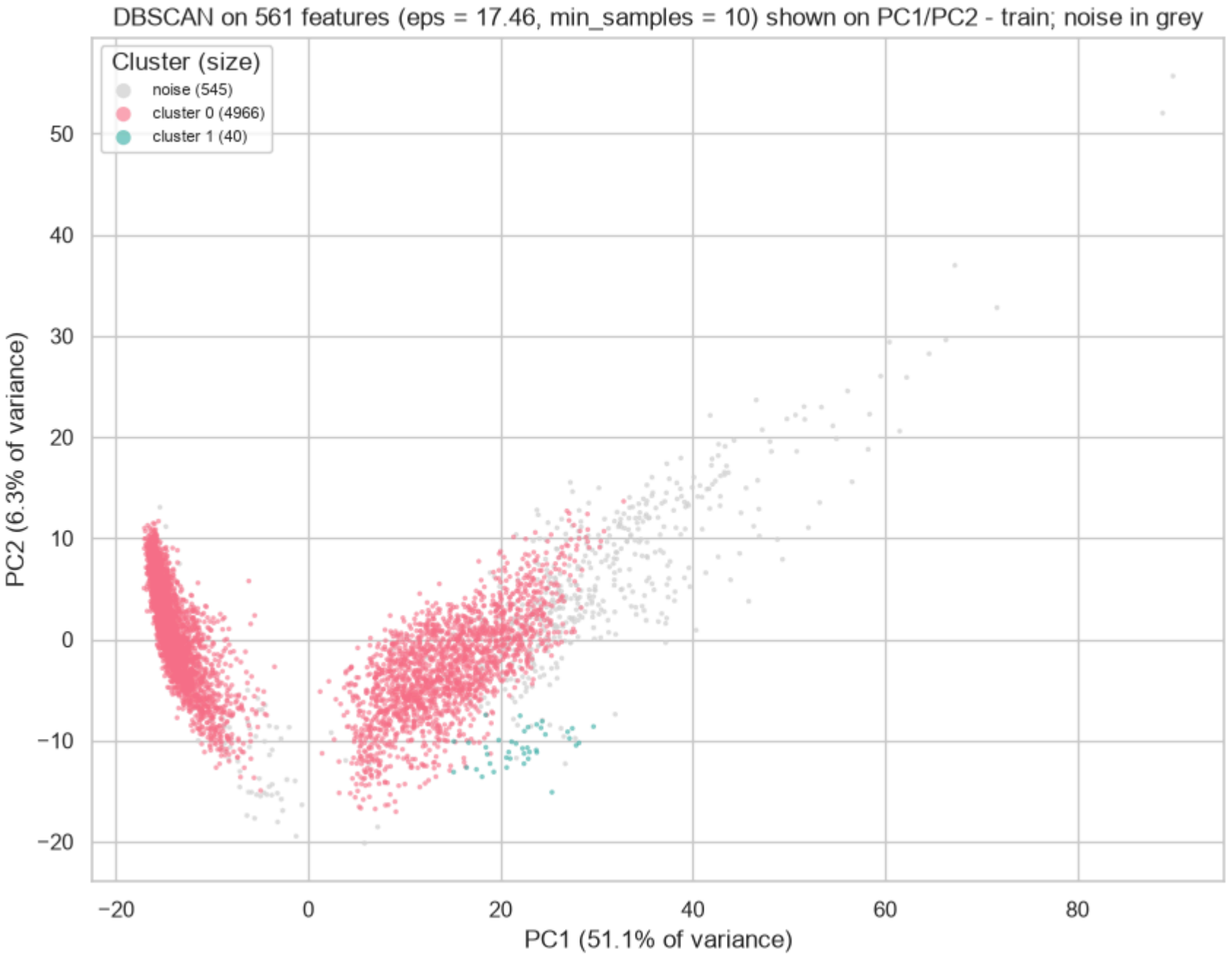
```
Selection: max silhouette with >= 2 clusters and noise <= 10%
-> eps = 17.456: 2 cluster(s), 9.8% noise, silhouette 0.3090, ARI 0.0266
```

In [28]:
```
lab_db561 = dbscan561_labels[eps_best561]
plot_clusters_pc12(lab_db561, f"DBSCAN on 561 features (eps = {eps_best561:.2f}, min_samples = {MIN_SAMPLES}) "
                              "shown on PC1/PC2 - train; noise in grey", "phase3_dbscan561_best_pc12")

dbscan_space_comparison = pd.DataFrame([
    {"run": f"first {k_elbow} PCA components (main)", "n_dims": k_elbow, "eps_knee": eps_knee, "eps_selected": eps_best,
     "rule_satisfied": True, "n_clusters": int(best_row["n_clusters"]), "noise_pct": best_row["noise_pct"],
     "silhouette": best_row["silhouette"], "ari": best_row["ari"], "ari_non_noise": best_row["ari_non_noise"]},
```

```
    {"run": "all 561 standardized features (comparison)", "n_dims": X_tr.shape[1], "eps_knee": eps_knee561,
     "eps_selected": eps_best561, "rule_satisfied": rule_ok561, "n_clusters": int(best561["n_clusters"]),
     "noise_pct": best561["noise_pct"], "silhouette": best561["silhouette"], "ari": best561["ari"],
     "ari_non_noise": best561["ari_non_noise"]},
])
save_table(dbscan_space_comparison, "phase3_dbscan_3pc_vs_561", index=False)
display(dbscan_space_comparison.round(4))
print("Silhouettes are computed in each run's own space (3-D vs 561-D), so compare ARI across runs, not silhouette.")
```

```
saved figures\phase3_dbscan561_best_pc12.png
```




saved results\phase3_dbscan_3pc_vs_561.csv

| | run | n_dims | eps_knee | eps_selected | rule_satisfied | n_clusters | noise_pct | silhouette | ari | ari_non_noise |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | first 3 PCA components (main) | 3 | 4.1108 | 3.0831 | True | 3 | 2.8643 | 0.6146 | 0.3271 | 0.3290 |
| **1** | all 561 standardized features (comparison) | 561 | 21.8194 | 17.4555 | True | 2 | 9.8181 | 0.3090 | 0.0266 | 0.0017 |

```
Silhouettes are computed in each run's own space (3-D vs 561-D), so compare ARI across runs, not silhouette.
```

## Report - 2. Clustering

**K-Means** ( `phase3_kmeans_silhouette_inertia_vs_k.png` ). We ran K-Means on all 561 standardized features for k = 2…10. Silhouette is highest at **k = 2 (0.407)**, stays fairly high at k = 3 (0.35), then collapses to 0.10–0.16 for k ≥ 4. Inertia decreases smoothly with no clear bend, so it gives little guidance, which is why we chose k by silhouette.

- **k = 2** ( `phase3_kmeans_best_pc12.png` , `phase3_kmeans_k2_contingency.png` ) recovers the moving/stationary split almost perfectly. One cluster holds 887/888 walking, all upstairs and all downstairs windows. The other holds all sitting, all standing and 1,071/1,076 laying windows. ARI against the true activities is 0.33: high purity, but only two groups for six labels.
- **k = 6** ( `phase3_kmeans_k6_contingency.png` ) matches the activities better (ARI 0.43) but its clusters are much less compact (silhouette 0.11). Laying becomes almost its own cluster (902 of its 967 windows are laying), but sitting and standing are mixed across two clusters, and the three walking activities are spread over three mixed clusters.

So silhouette rewards the obvious two-way split, and only forcing k = 6 begins to recover individual activities. The cluster structure in the feature space is coarser than the label structure.

**DBSCAN** ( `phase3_dbscan_k_distance.png` , `phase3_dbscan_eps_sweep.png` , `phase3_dbscan_best_pc12.png` ). We ran DBSCAN (min_samples = 10) on the first k = 3 PCA components, where distances are meaningful for a density method. The k-distance plot has its knee at eps ≈ 4.1. We swept eps from 2.1 to 8.2 and selected the setting with the highest silhouette among those labeling at most 10% of points as noise. The noise cap stops DBSCAN from inflating its silhouette by discarding hard, in-between points.

- **Selected eps = 3.08**: 3 clusters, 2.9% noise, silhouette 0.61, ARI 0.33. One cluster contains all three static activities, one contains walking, upstairs and downstairs, and a small cluster of 44 windows is entirely walking upstairs (the separated tail at PC2 ≈ −10). 115 of the 159 noise points are walking downstairs, the high-intensity tail visible in the PCA plot.
- **Effect of eps.** Small eps (≈2–2.5) fragments the data into 5–6 clusters with up to 8% noise. At eps ≈ 2.5 DBSCAN even finds 6 clusters, but with lower silhouette and more noise. Between 3 and 4.7 the result is mostly the same 3-cluster solution (briefly 4 near eps ≈ 3.5–3.7). From eps ≈ 4.9 upward everything merges into a single cluster and ARI drops to 0.

**DBSCAN on all 561 features** ( `phase3_dbscan561_k_distance.png` , `phase3_dbscan561_best_pc12.png` ). For comparison we repeated DBSCAN on the full feature set. The k-distance knee moves to eps ≈ 21.8, because every distance is large in 561 dimensions. No setting is good. Any eps that separates moving from stationary (≈14–16) discards 15–30% of points as noise, and the first setting under the 10% noise cap (eps = 17.46) merges almost everything into one cluster (ARI 0.03). This supports running the main DBSCAN on the PCA components.

**Comparison.** Both methods agree on the dominant structure, moving vs stationary, and neither separates the three static postures without labels. K-Means needs k chosen in advance and assigns every point. DBSCAN finds the number of clusters itself, isolates the downstairs intensity tail as noise, and picks out the walking-upstairs subgroup, but it is very sensitive to eps and only works well in a low-dimensional space.

# Phase 4 - Two ways to shrink features

The cells below compare two ways of reducing the 561 features, using the **train split only** (dev and test are not touched):

- **TRANSFORM** - PCA, reusing the Phase 2 fit on train. Train is projected onto the first `k_elbow` , `k90` and `k95` components (elbow, 90% and 95% cumulative variance).
- **SELECT** - keep a subset of the original, interpretable features. A `RandomForestClassifier` ( `n_estimators=300, random_state=42, n_jobs=-1` ) is fitted on train and the 561 features are ranked by impurity importance. The top 20 are summarised by domain, signal and axis, and cumulative importance shows how many features carry 50/80/90% of the total.

Labels are used only to fit the random forest on train. Caveat: impurity importance is shared among correlated features, and Phase 2 showed many near-duplicates, so one useful signal can rank lower than it deserves when several copies split its credit.

## 4.1 TRANSFORM: PCA (Phase 2 fit on train)

In [29]:
```
assert pca.n_features_in_ == X_tr.shape[1]      # same PCA object fitted on train in Phase 2
Z_all_tr = pca.transform(X_tr)

PCA_SETS = {"pca_elbow": k_elbow, "pca_90": k90, "pca_95": k95}
pca_reduced_tr = {name: Z_all_tr[:, :k] for name, k in PCA_SETS.items()}

pca_reduced_summary = pd.DataFrame([
    {"set": name, "n_components": k, "train_shape": str(pca_reduced_tr[name].shape),
     "retained_variance": cum_evr[k - 1], "reduction_vs_561": f"{100 * (1 - k / X_tr.shape[1]):.1f}%"}
    for name, k in PCA_SETS.items()
])
```

```
save_table(pca_reduced_summary, "phase4_pca_reduced_sets", index=False)
display(pca_reduced_summary.round(4))
```

```
saved results\phase4_pca_reduced_sets.csv
```

| | set | n_components | train_shape | retained_variance | reduction_vs_561 |
|---|---|---|---|---|---|
| **0** | pca_elbow | 3 | (5551, 3) | 0.6016 | 99.5% |
| **1** | pca_90 | 61 | (5551, 61) | 0.9003 | 89.1% |
| **2** | pca_95 | 100 | (5551, 100) | 0.9507 | 82.2% |

## 4.2 SELECT: random-forest impurity importance

In [30]:

```
from sklearn.ensemble import RandomForestClassifier
import re

t0 = time.perf_counter()
rf = RandomForestClassifier(n_estimators=300, random_state=RANDOM_STATE, n_jobs=-1).fit(X_tr, y_tr_arr)
rf_fit_seconds = time.perf_counter() - t0
record_timing("phase4", "random_forest_fit", rf_fit_seconds)
print(f"Random forest (300 trees) fitted on train in {rf_fit_seconds:.1f}s")


def feature_groups(name):
    """(domain, signal, axis) for a feature name.

    domain: time / frequency / angle.  signal: body acc / gravity acc / gyroscope / angle.
    axis: X / Y / Z, an axis pair for correlation features (e.g. 'X,Y'), or 'none' (magnitude features).
    features.txt does not name the axis of bandsEnergy features: each signal lists the same 14 bands three
    times, in X, Y, Z order (the convention used by all other features). Phase 1 made those names unique with
    suffixes, so no suffix = X, '_2' = Y, '_3' = Z.
    """
    if name.startswith("angle("):
        m = re.match(r"angle\(([XYZ]),", name)
        return "angle", "angle", (m.group(1) if m else "none")
    domain = "time" if name[0] == "t" else "frequency"
    sig = name.split("-", 1)[0]
    signal = "gyroscope" if "Gyro" in sig else "gravity acc" if "Gravity" in sig else "body acc"
    if "bandsEnergy" in name:
        axis = {"_2": "Y", "_3": "Z"}.get(name[-2:], "X")
    else:
        m = re.search(r"-([XYZ])(?:,([XYZ]))?(?:,\d)?$", name)
        axis = (m.group(1) + ("," + m.group(2) if m.group(2) else "")) if m else "none"
```

```
    return domain, signal, axis


importance = (pd.DataFrame({"feature": FEATURE_NAMES, "importance": rf.feature_importances_})
              .sort_values("importance", ascending=False, ignore_index=True))
importance.insert(0, "rank", np.arange(1, len(importance) + 1))
importance["cumulative_importance"] = importance["importance"].cumsum()
importance[["domain", "signal", "axis"]] = pd.DataFrame(importance["feature"].map(feature_groups).tolist())

save_table(importance, "phase4_rf_feature_importance_ranking", index=False)
display(importance.head(20).round(4))
```

```
Random forest (300 trees) fitted on train in 1.5s
saved results\phase4_rf_feature_importance_ranking.csv
```

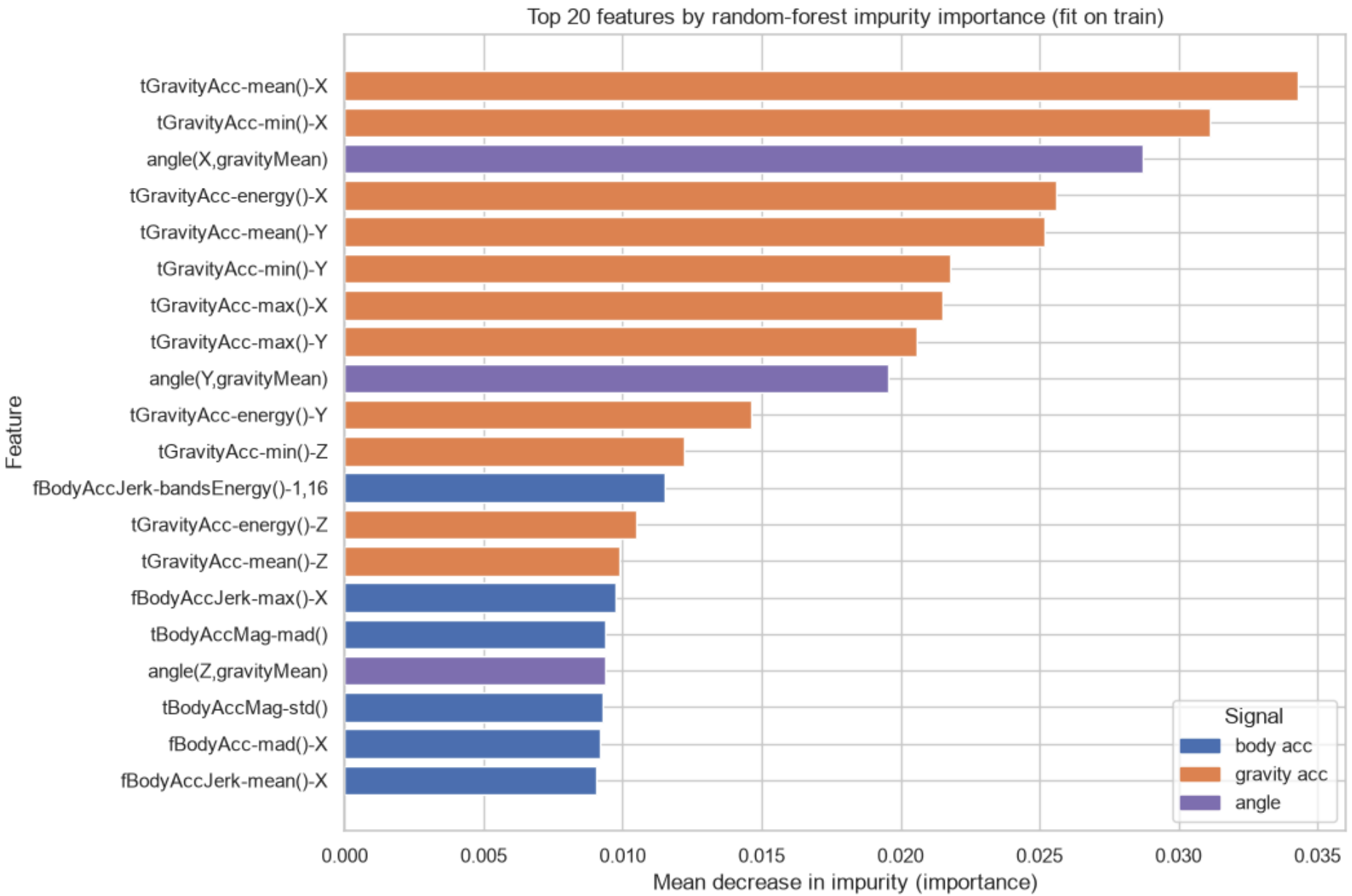
| | rank | feature | importance | cumulative_importance | domain | signal | axis |
|---|---|---|---|---|---|---|---|
| **0** | 1 | tGravityAcc-mean()-X | 0.0343 | 0.0343 | time | gravity acc | X |
| **1** | 2 | tGravityAcc-min()-X | 0.0311 | 0.0654 | time | gravity acc | X |
| **2** | 3 | angle(X,gravityMean) | 0.0287 | 0.0942 | angle | angle | X |
| **3** | 4 | tGravityAcc-energy()-X | 0.0256 | 0.1198 | time | gravity acc | X |
| **4** | 5 | tGravityAcc-mean()-Y | 0.0252 | 0.1449 | time | gravity acc | Y |
| **5** | 6 | tGravityAcc-min()-Y | 0.0218 | 0.1667 | time | gravity acc | Y |
| **6** | 7 | tGravityAcc-max()-X | 0.0215 | 0.1883 | time | gravity acc | X |
| **7** | 8 | tGravityAcc-max()-Y | 0.0206 | 0.2088 | time | gravity acc | Y |
| **8** | 9 | angle(Y,gravityMean) | 0.0196 | 0.2284 | angle | angle | Y |
| **9** | 10 | tGravityAcc-energy()-Y | 0.0146 | 0.2430 | time | gravity acc | Y |
| **10** | 11 | tGravityAcc-min()-Z | 0.0122 | 0.2552 | time | gravity acc | Z |
| **11** | 12 | fBodyAccJerk-bandsEnergy()-1,16 | 0.0115 | 0.2668 | frequency | body acc | X |
| **12** | 13 | tGravityAcc-energy()-Z | 0.0105 | 0.2772 | time | gravity acc | Z |
| **13** | 14 | tGravityAcc-mean()-Z | 0.0099 | 0.2871 | time | gravity acc | Z |
| **14** | 15 | fBodyAccJerk-max()-X | 0.0097 | 0.2969 | frequency | body acc | X |
| **15** | 16 | tBodyAccMag-mad() | 0.0094 | 0.3063 | time | body acc | none |
| **16** | 17 | angle(Z,gravityMean) | 0.0094 | 0.3156 | angle | angle | Z |
| **17** | 18 | tBodyAccMag-std() | 0.0093 | 0.3249 | time | body acc | none |
| **18** | 19 | fBodyAcc-mad()-X | 0.0092 | 0.3341 | frequency | body acc | X |
| **19** | 20 | fBodyAccJerk-mean()-X | 0.0090 | 0.3431 | frequency | body acc | X |

In [31]:
```
TOP_N = 20
top = importance.head(TOP_N)
signal_colors = {"body acc": "#4C72B0", "gravity acc": "#DD8452", "gyroscope": "#55A868", "angle": "#8172B2"}

fig, ax = plt.subplots(figsize=(10, 8))
ax.barh(top["feature"], top["importance"], color=top["signal"].map(signal_colors))
ax.invert_yaxis()
```

```
ax.set_title(f"Top {TOP_N} features by random-forest impurity importance (fit on train)")
ax.set_xlabel("Mean decrease in impurity (importance)")
ax.set_ylabel("Feature")
from matplotlib.patches import Patch
ax.legend(handles=[Patch(color=c, label=s) for s, c in signal_colors.items() if s in set(top["signal"])],
          title="Signal", loc="lower right")
save_fig(fig, "phase4_rf_top20_importance")
plt.show()
```

```
saved figures\phase4_rf_top20_importance.png
```




In [32]:
```
# Summarise the top 20 along three separate dimensions (each breakdown sums to 20).
# Every group that exists among the 561 features is listed, so groups absent from the top 20 show as 0.
group_rows = []
for dim in ["domain", "signal", "axis"]:
```

```
    counts = top[dim].value_counts().reindex(importance[dim].unique(), fill_value=0)
    counts = counts.sort_values(ascending=False, kind="stable")
    print(f"Top {TOP_N} by {dim}: " + ", ".join(f"{k} = {v}" for k, v in counts.items()))
    group_rows += [{"dimension": dim, "group": k, "count": int(v),
                    "share_of_top20_importance": top.loc[top[dim] == k, "importance"].sum() / top["importance"].sum()}
                   for k, v in counts.items()]
top20_groups = pd.DataFrame(group_rows)
save_table(top20_groups, "phase4_top20_group_counts", index=False)
display(top20_groups.round(3))
```

```
Top 20 by domain: time = 13, frequency = 4, angle = 3
Top 20 by signal: gravity acc = 11, body acc = 6, angle = 3, gyroscope = 0
Top 20 by axis: X = 9, Y = 5, Z = 4, none = 2, X,Y = 0, Y,Z = 0, X,Z = 0
saved results\phase4_top20_group_counts.csv
```

| | dimension | group | count | share_of_top20_importance |
|---|---|---|---|---|
| **0** | domain | time | 13 | 0.717 |
| **1** | domain | frequency | 4 | 0.115 |
| **2** | domain | angle | 3 | 0.168 |
| **3** | signal | gravity acc | 11 | 0.662 |
| **4** | signal | body acc | 6 | 0.170 |
| **5** | signal | angle | 3 | 0.168 |
| **6** | signal | gyroscope | 0 | 0.000 |
| **7** | axis | X | 9 | 0.527 |
| **8** | axis | Y | 5 | 0.297 |
| **9** | axis | Z | 4 | 0.122 |
| **10** | axis | none | 2 | 0.054 |
| **11** | axis | X,Y | 0 | 0.000 |
| **12** | axis | Y,Z | 0 | 0.000 |
| **13** | axis | X,Z | 0 | 0.000 |

## 4.3 Cumulative importance vs number of features

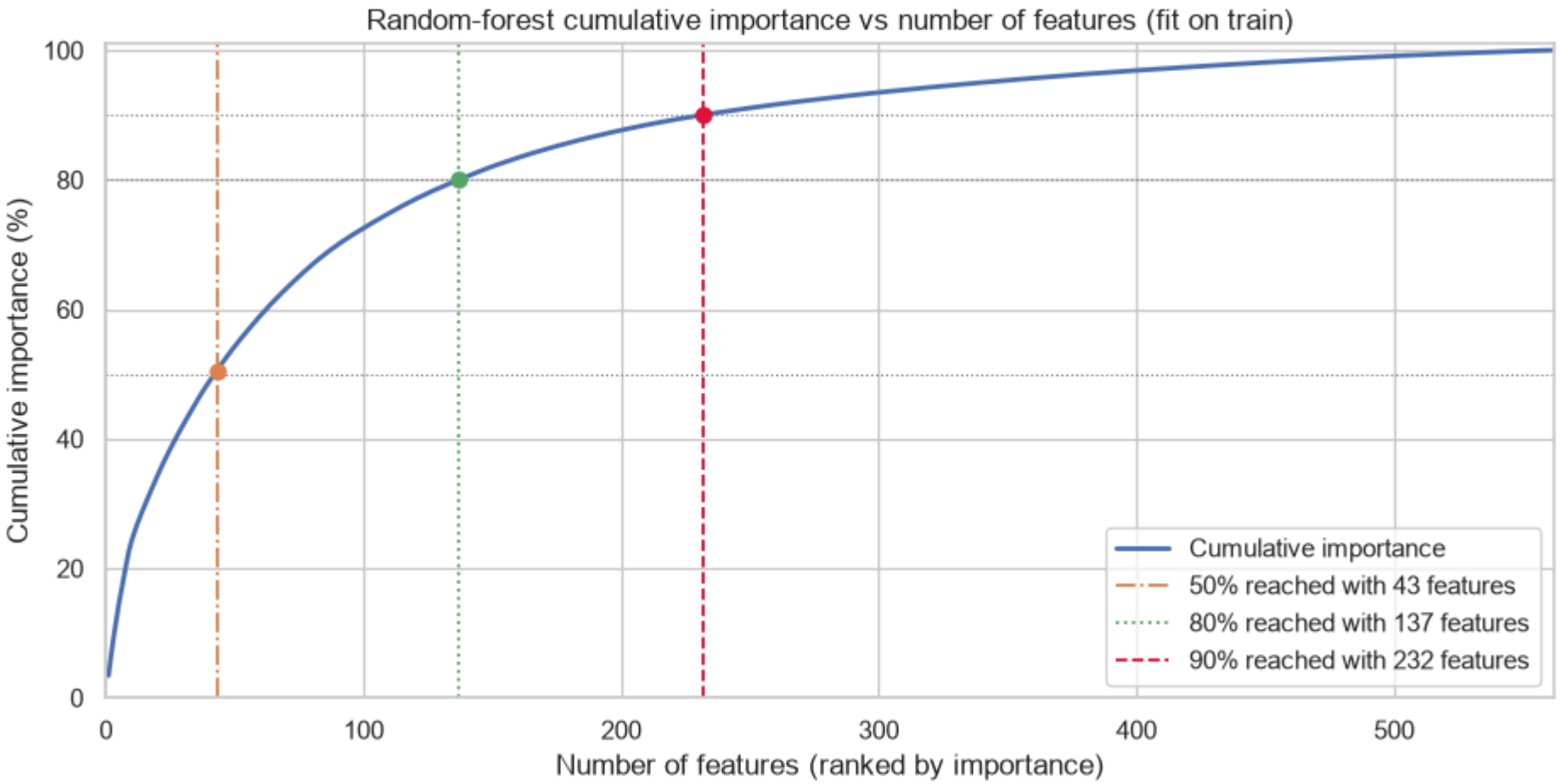
In [33]:
```
thresholds = [0.50, 0.80, 0.90]
n_for = {t: int(np.searchsorted(importance["cumulative_importance"].to_numpy(), t) + 1) for t in thresholds}
for t in thresholds:
    print(f"{int(t * 100)}% of total importance reached with {n_for[t]} features "
          f"({100 * n_for[t] / len(importance):.1f}% of 561)")

importance_thresholds = pd.DataFrame({"importance_threshold": thresholds,
                                      "n_features": [n_for[t] for t in thresholds],
                                      "pct_of_561": [round(100 * n_for[t] / len(importance), 1) for t in thresholds]})
```

```
save_table(importance_thresholds, "phase4_importance_thresholds", index=False)

fig, ax = plt.subplots(figsize=(11, 5))
ax.plot(importance["rank"], 100 * importance["cumulative_importance"], lw=2, color="#4C72B0",
        label="Cumulative importance")
for t, color, ls in zip(thresholds, ["#DD8452", "#55A868", "crimson"], ["-.", ":", "--"]):
    ax.axvline(n_for[t], color=color, ls=ls, lw=1.3, label=f"{int(t * 100)}% reached with {n_for[t]} features")
    ax.axhline(100 * t, color="gray", lw=0.8, ls=":")
    ax.scatter([n_for[t]], [100 * importance["cumulative_importance"].iloc[n_for[t] - 1]], color=color, s=40, zorder=5)
ax.set_title("Random-forest cumulative importance vs number of features (fit on train)")
ax.set_xlabel("Number of features (ranked by importance)")
ax.set_ylabel("Cumulative importance (%)")
ax.set_xlim(0, len(importance) + 1)
ax.set_ylim(0, 101)
ax.legend(loc="lower right")
save_fig(fig, "phase4_rf_cumulative_importance")
plt.show()
```

```
50% of total importance reached with 43 features (7.7% of 561)
80% of total importance reached with 137 features (24.4% of 561)
90% of total importance reached with 232 features (41.4% of 561)
saved results\phase4_importance_thresholds.csv
saved figures\phase4_rf_cumulative_importance.png
```




## Phase 5 - Classification with fewer features

The cells below train the same classifier, `LogisticRegression(max_iter=5000, random_state=42)`, on train under three feature conditions and compare them on **dev**:

- **full** - all 561 standardized features (baseline);
- **PCA** - the top-d principal components (Phase 2 PCA, fit on train);
- **RF selection** - the top-d original features of the Phase 4 random-forest ranking (fit on train).

d is swept over 2...300. For each approach the reduced model is the **smallest d whose dev accuracy is within 1 percentage point of the full baseline** (if none reaches it, the d with the best dev accuracy, stated explicitly). The chosen models are then compared on dev (confusion matrices) and on efficiency (training and prediction time, memory, one-time reduction cost). Finally - the only time a model is trained or evaluated with test data - the three models are retrained on train and evaluated once on test.

In [34]:
```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (accuracy_score, f1_score, precision_recall_fscore_support,
                             confusion_matrix, classification_report)

# Dev split from the Phase 1 cache (test arrays are dropped here and loaded only in 5.6)
dev_data = {k: v for k, v in load_cache().items() if not k.endswith("_test")}
X_dv = dev_data["X_dev"]
y_dv_arr = dev_data["y_dev"]
del dev_data
print(f"train: {X_tr.shape}, dev: {X_dv.shape}")


def make_clf():
    """The single classifier used for every condition."""
    return LogisticRegression(max_iter=5000, random_state=RANDOM_STATE)


# Column order of the Phase 4 random-forest ranking (most important first)
RF_ORDER = np.array([FEATURE_NAMES.index(f) for f in importance["feature"]])


def reduce_features(method, d, X):
    """Map standardized 561-D features to the representation of a condition."""
    if method == "full":
        return X
    if method == "pca":
        return pca.transform(X)[:, :d]          # Phase 2 PCA, fit on train only
    if method == "rf":
        return X[:, RF_ORDER[:d]]               # Phase 4 RF ranking, fit on train only
    raise ValueError(method)
```

```
train: (5551, 561), dev: (1801, 561)
```

## 5.1 Full baseline (561 features) on dev

In [35]:
```
t0 = time.perf_counter()
clf_full = make_clf().fit(X_tr, y_tr_arr)
record_timing("phase5", "logreg_fit_full_561", time.perf_counter() - t0)
pred_full_dv = clf_full.predict(X_dv)
base_acc = accuracy_score(y_dv_arr, pred_full_dv)
base_f1 = f1_score(y_dv_arr, pred_full_dv, average="macro")
print(f"Full baseline (561 features): dev accuracy = {base_acc:.4f}, dev macro F1 = {base_f1:.4f}")
```

```
Full baseline (561 features): dev accuracy = 0.9572, dev macro F1 = 0.9534
```

## 5.2 Sweep d for PCA and RF selection

In [36]:
```
D_GRID = [2, 3, 5, 9, 10, 20, 30, 50, 61, 75, 100, 150, 200, 300]
pca_tr_all, pca_dv_all = pca.transform(X_tr), pca.transform(X_dv)   # project once, slice per d

rows = []
for method in ["pca", "rf"]:
    for d in D_GRID:
        if method == "pca":
            A_tr, A_dv = pca_tr_all[:, :d], pca_dv_all[:, :d]
        else:
            A_tr, A_dv = X_tr[:, RF_ORDER[:d]], X_dv[:, RF_ORDER[:d]]
        t0 = time.perf_counter()
        clf = make_clf().fit(A_tr, y_tr_arr)
        record_timing("phase5", f"logreg_fit_{method}_d{d}", time.perf_counter() - t0)
        pred = clf.predict(A_dv)
        rows.append({"method": method, "d": d, "dev_accuracy": accuracy_score(y_dv_arr, pred),
                     "dev_macro_f1": f1_score(y_dv_arr, pred, average="macro")})
sweep = pd.DataFrame(rows)
sweep["acc_gap_vs_full_pp"] = 100 * (sweep["dev_accuracy"] - base_acc)
sweep = pd.concat([pd.DataFrame([{"method": "full", "d": 561, "dev_accuracy": base_acc, "dev_macro_f1": base_f1,
                                  "acc_gap_vs_full_pp": 0.0}]), sweep], ignore_index=True)
save_table(sweep, "phase5_dev_sweep", index=False)
display(sweep.pivot(index="d", columns="method", values="dev_accuracy").round(4))
```

```
saved results\phase5_dev_sweep.csv
```

| method<br>d | full | pca | rf |
|---|---|---|---|
| **2** | NaN | 0.5386 | 0.4181 |
| **3** | NaN | 0.7146 | 0.4814 |
| **5** | NaN | 0.8323 | 0.6546 |
| **9** | NaN | 0.8756 | 0.6657 |
| **10** | NaN | 0.8701 | 0.6696 |
| **20** | NaN | 0.8901 | 0.8923 |
| **30** | NaN | 0.9145 | 0.9328 |
| **50** | NaN | 0.9284 | 0.9473 |
| **61** | NaN | 0.9317 | 0.9550 |
| **75** | NaN | 0.9434 | 0.9611 |
| **100** | NaN | 0.9461 | 0.9606 |
| **150** | NaN | 0.9545 | 0.9511 |
| **200** | NaN | 0.9595 | 0.9567 |
| **300** | NaN | 0.9600 | 0.9550 |
| **561** | 0.9572 | NaN | NaN |

## 5.3 Choosing the reduced models (smallest d within 1 pp of the full baseline)

In [37]:
```
TOL = 0.01   # 1 percentage point


def choose_d(method):
    s = sweep[sweep["method"] == method]
    ok = s[s["dev_accuracy"] >= base_acc - TOL]
    if len(ok):
        row = ok.loc[ok["d"].idxmin()]
        return int(row["d"]), True, row
    row = s.loc[s["dev_accuracy"].idxmax()]
    return int(row["d"]), False, row


chosen = {}
for method, label in [("pca", "PCA"), ("rf", "RF selection")]:
    d, ok, row = choose_d(method)
    chosen[method] = d
    if ok:
        print(f"{label}: chosen d = {d} (smallest d within 1 pp of full; dev accuracy {row['dev_accuracy']:.4f}, "
              f"{row['acc_gap_vs_full_pp']:+.2f} pp vs full)")
    else:
```

```
        print(f"{label}: NO d in the grid reaches within 1 pp of the full baseline -> using the best-accuracy d = {d} "
              f"(dev accuracy {row['dev_accuracy']:.4f}, {row['acc_gap_vs_full_pp']:+.2f} pp vs full)")
d_pca, d_rf = chosen["pca"], chosen["rf"]

chosen_table = pd.DataFrame([
    {"model": "full", "method": "full", "d": 561, "dev_accuracy": base_acc, "dev_macro_f1": base_f1},
    *[{"model": f"{m}_d{chosen[m]}", "method": m, "d": chosen[m],
       **sweep[(sweep["method"] == m) & (sweep["d"] == chosen[m])][["dev_accuracy", "dev_macro_f1"]].iloc[0].to_dict()}
      for m in ["pca", "rf"]],
])
chosen_table["within_1pp_of_full"] = chosen_table["dev_accuracy"] >= base_acc - TOL
save_table(chosen_table, "phase5_chosen_models", index=False)
display(chosen_table.round(4))
```

```
PCA: chosen d = 150 (smallest d within 1 pp of full; dev accuracy 0.9545, -0.28 pp vs full)
RF selection: chosen d = 50 (smallest d within 1 pp of full; dev accuracy 0.9473, -1.00 pp vs full)
saved results\phase5_chosen_models.csv
```

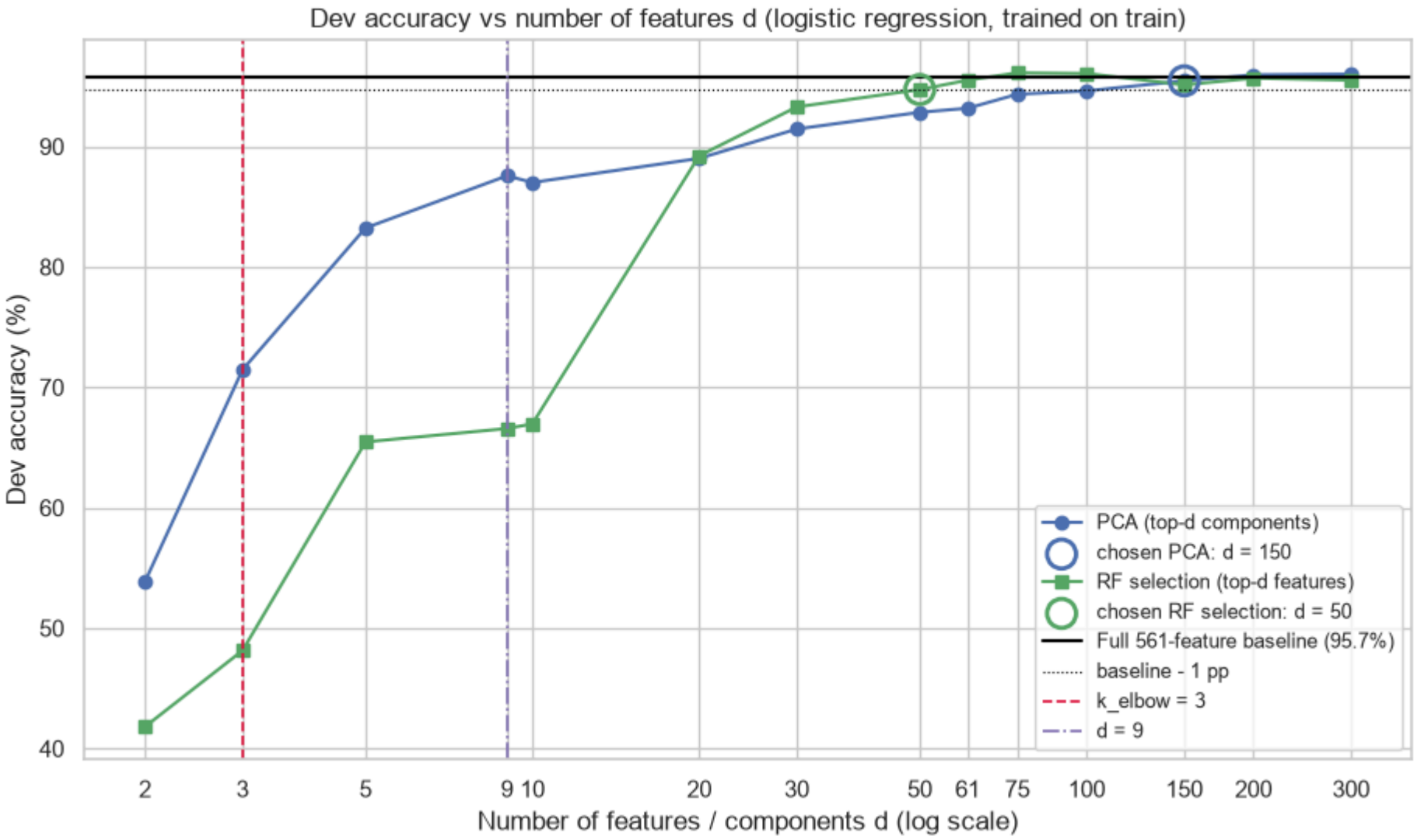
| | model | method | d | dev_accuracy | dev_macro_f1 | within_1pp_of_full |
|---|---|---|---|---|---|---|
| **0** | full | full | 561 | 0.9572 | 0.9534 | True |
| **1** | pca_d150 | pca | 150 | 0.9545 | 0.9497 | True |
| **2** | rf_d50 | rf | 50 | 0.9473 | 0.9421 | True |

In [38]:

```
fig, ax = plt.subplots(figsize=(11, 6))
style = {"pca": ("#4C72B0", "o", "PCA (top-d components)"), "rf": ("#55A868", "s", "RF selection (top-d features)")}
for m, (color, marker, label) in style.items():
    s = sweep[sweep["method"] == m]
    ax.plot(s["d"], 100 * s["dev_accuracy"], marker=marker, color=color, label=label)
    sel = s[s["d"] == chosen[m]]
    ax.scatter(sel["d"], 100 * sel["dev_accuracy"], s=180, facecolor="none", edgecolor=color, lw=2, zorder=5,
               label=f"chosen {label.split(' (')[0]}: d = {chosen[m]}")
ax.axhline(100 * base_acc, color="black", lw=1.5, label=f"Full 561-feature baseline ({100 * base_acc:.1f}%)")
ax.axhline(100 * (base_acc - TOL), color="black", lw=0.8, ls=":", label="baseline - 1 pp")
ax.axvline(k_elbow, color="crimson", ls="--", lw=1.2, label=f"k_elbow = {k_elbow}")
ax.axvline(9, color="#8172B2", ls="-.", lw=1.2, label="d = 9")
ax.set_xscale("log")
ax.set_xticks(D_GRID)
ax.set_xticklabels(D_GRID)
ax.minorticks_off()
ax.set_title("Dev accuracy vs number of features d (logistic regression, trained on train)")
ax.set_xlabel("Number of features / components d (log scale)")
ax.set_ylabel("Dev accuracy (%)")
ax.legend(loc="lower right", fontsize=9)
```

```
save_fig(fig, "phase5_dev_accuracy_vs_d")
plt.show()
```

```
saved figures\phase5_dev_accuracy_vs_d.png
```




## 5.4 Confusion matrices on dev (row-normalized)

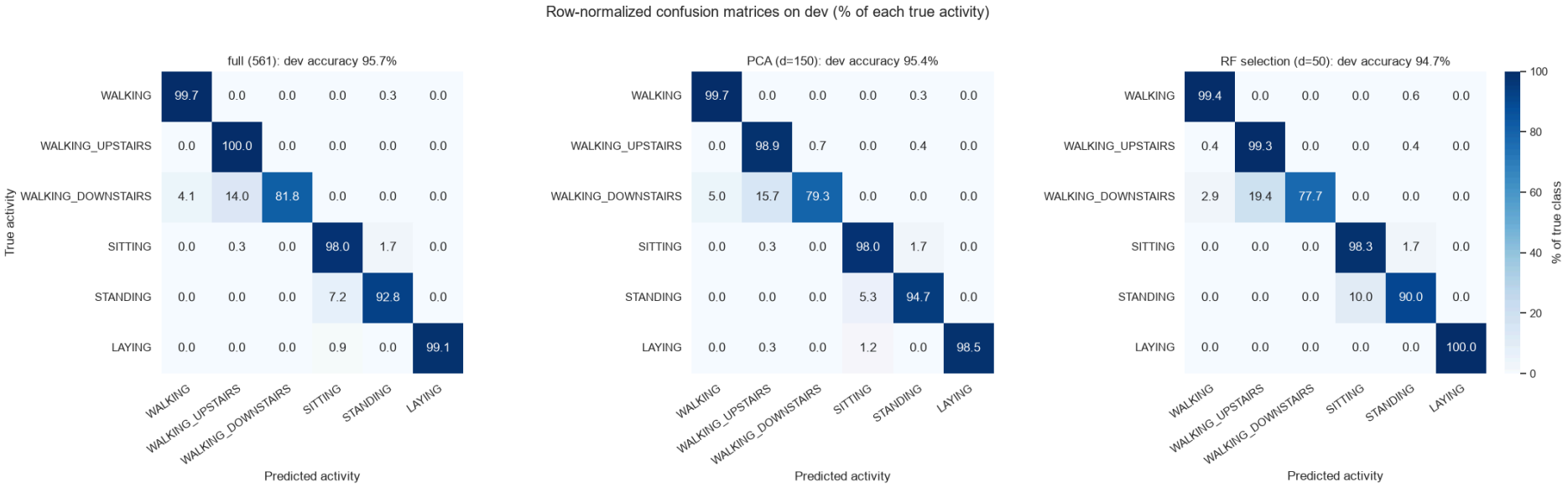
In [39]:
```
MODELS = {"full (561)": ("full", 561), f"PCA (d={d_pca})": ("pca", d_pca), f"RF selection (d={d_rf})": ("rf", d_rf)}
```

```
dev_preds = {}
for name, (m, d) in MODELS.items():
    clf = clf_full if m == "full" else make_clf().fit(reduce_features(m, d, X_tr), y_tr_arr)
    dev_preds[name] = clf.predict(reduce_features(m, d, X_dv))

fig, axes = plt.subplots(1, 3, figsize=(22, 6.5))
top_pairs = {}
for ax, (name, pred) in zip(axes, dev_preds.items()):
    cm = confusion_matrix(y_dv_arr, pred, labels=ACTIVITY_ORDER, normalize="true")
    cm_df = pd.DataFrame(cm, index=ACTIVITY_ORDER, columns=ACTIVITY_ORDER)
    save_table(cm_df.round(4), "phase5_dev_confusion_" + name.split(" ")[0].lower())
    sns.heatmap(100 * cm_df, annot=True, fmt=".1f", cmap="Blues", vmin=0, vmax=100, cbar=ax is axes[-1],
                cbar_kws={"label": "% of true class"}, ax=ax, square=True)
    ax.set_title(f"{name}: dev accuracy {100 * accuracy_score(y_dv_arr, pred):.1f}%")
    ax.set_xlabel("Predicted activity")
    ax.set_ylabel("True activity" if ax is axes[0] else "")
    plt.setp(ax.get_xticklabels(), rotation=35, ha="right", rotation_mode="anchor")
    # Unordered activity pairs ranked by total confusion (A->B rate + B->A rate)
    pairs = [((a, b), cm_df.loc[a, b] + cm_df.loc[b, a], cm_df.loc[a, b], cm_df.loc[b, a])
             for i, a in enumerate(ACTIVITY_ORDER) for b in ACTIVITY_ORDER[i + 1:]]
    top_pairs[name] = sorted(pairs, key=lambda t: -t[1])[:3]
fig.suptitle("Row-normalized confusion matrices on dev (% of each true activity)", y=1.02)
fig.tight_layout()
save_fig(fig, "phase5_dev_confusion_matrices")
plt.show()

for name, pairs in top_pairs.items():
    print(f"{name} - most confused pairs:")
    for (a, b), total, ab, ba in pairs:
        print(f"    {a} <-> {b}: {100 * ab:.1f}% of {a} -> {b}, {100 * ba:.1f}% of {b} -> {a}")
common = set.intersection(*[{p[0] for p in pairs} for pairs in top_pairs.values()])
top1 = {pairs[0][0] for pairs in top_pairs.values()}
print("\nSame across all three models:")
print("  top-3 confused pairs shared by all:", sorted(f"{a} <-> {b}" for a, b in common) or "none")
print("  most confused pair:", (" / ".join(f"{a} <-> {b}" for a, b in top1)) if len(top1) == 1
      else f"differs between models: {sorted(f'{a} <-> {b}' for a, b in top1)}")
moving, still = ACTIVITY_ORDER[:3], ACTIVITY_ORDER[3:]
cross = {name: float(100 * np.mean([(t in moving) != (p in moving) for t, p in zip(y_dv_arr, pred)]))
         for name, pred in dev_preds.items()}
print("  moving <-> stationary errors (% of dev windows):", {k: round(v, 2) for k, v in cross.items()})
```

```
saved results\phase5_dev_confusion_full.csv
saved results\phase5_dev_confusion_pca.csv
saved results\phase5_dev_confusion_rf.csv
saved figures\phase5_dev_confusion_matrices.png
```



```
full (561) - most confused pairs:
    WALKING_UPSTAIRS <-> WALKING_DOWNSTAIRS: 0.0% of WALKING_UPSTAIRS -> WALKING_DOWNSTAIRS, 14.0% of WALKING_DOWNSTAIRS -> WALKIN
G_UPSTAIRS
    SITTING <-> STANDING: 1.7% of SITTING -> STANDING, 7.2% of STANDING -> SITTING
    WALKING <-> WALKING_DOWNSTAIRS: 0.0% of WALKING -> WALKING_DOWNSTAIRS, 4.1% of WALKING_DOWNSTAIRS -> WALKING
PCA (d=150) - most confused pairs:
    WALKING_UPSTAIRS <-> WALKING_DOWNSTAIRS: 0.7% of WALKING_UPSTAIRS -> WALKING_DOWNSTAIRS, 15.7% of WALKING_DOWNSTAIRS -> WALKIN
G_UPSTAIRS
    SITTING <-> STANDING: 1.7% of SITTING -> STANDING, 5.3% of STANDING -> SITTING
    WALKING <-> WALKING_DOWNSTAIRS: 0.0% of WALKING -> WALKING_DOWNSTAIRS, 5.0% of WALKING_DOWNSTAIRS -> WALKING
RF selection (d=50) - most confused pairs:
    WALKING_UPSTAIRS <-> WALKING_DOWNSTAIRS: 0.0% of WALKING_UPSTAIRS -> WALKING_DOWNSTAIRS, 19.4% of WALKING_DOWNSTAIRS -> WALKIN
G_UPSTAIRS
    SITTING <-> STANDING: 1.7% of SITTING -> STANDING, 10.0% of STANDING -> SITTING
    WALKING <-> WALKING_DOWNSTAIRS: 0.0% of WALKING -> WALKING_DOWNSTAIRS, 2.9% of WALKING_DOWNSTAIRS -> WALKING

Same across all three models:
  top-3 confused pairs shared by all: ['SITTING <-> STANDING', 'WALKING <-> WALKING_DOWNSTAIRS', 'WALKING_UPSTAIRS <-> WALKING_DOW
NSTAIRS']
  most confused pair: WALKING_UPSTAIRS <-> WALKING_DOWNSTAIRS
  moving <-> stationary errors (% of dev windows): {'full (561)': 0.11, 'PCA (d=150)': 0.22, 'RF selection (d=50)': 0.17}
```

## 5.5 Efficiency: full vs chosen reduced models

Training time = median of 5 fits on train; prediction time = median of 5 `predict` calls on the already-reduced dev matrix; `reduce_dev_seconds` = median time to turn the 561 standardized dev features into that matrix (PCA projection or column selection). Memory = size of the feature matrices. One-time reduction cost = median of 5 fits of the reducer on train ( `PCA(n_components=d)` or the 300-tree random forest).

In [40]:
```python
from sklearn.ensemble import RandomForestClassifier


def median_time(fn, n=5):
    ts = []
    for _ in range(n):
        t0 = time.perf_counter()
        fn()
        ts.append(time.perf_counter() - t0)
    return float(np.median(ts))
```

```
reducer_cost = {
    "full": 0.0,
    "pca": median_time(lambda: PCA(n_components=d_pca, random_state=RANDOM_STATE).fit(X_tr)),
    "rf": median_time(lambda: RandomForestClassifier(n_estimators=300, random_state=RANDOM_STATE, n_jobs=-1)
                      .fit(X_tr, y_tr_arr)),
}

eff_rows = []
for name, (m, d) in MODELS.items():
    A_tr, A_dv = reduce_features(m, d, X_tr), reduce_features(m, d, X_dv)
    clf = make_clf().fit(A_tr, y_tr_arr)
    eff_rows.append({
        "model": name, "method": m, "d": d,
        "train_seconds": median_time(lambda: make_clf().fit(A_tr, y_tr_arr)),
        "predict_dev_seconds": median_time(lambda: clf.predict(A_dv)),
        "reduce_dev_seconds": median_time(lambda: reduce_features(m, d, X_dv)),
        "train_matrix_mb": A_tr.nbytes / 1e6, "dev_matrix_mb": A_dv.nbytes / 1e6,
        "one_time_reduction_fit_seconds": reducer_cost[m],
    })
efficiency = pd.DataFrame(eff_rows)
for col in ["train_seconds", "predict_dev_seconds"]:   # no ratio for reduce_dev_seconds: full has no reduction step
    efficiency[f"{col}_vs_full"] = efficiency[col] / efficiency.loc[0, col]
save_table(efficiency, "phase5_efficiency", index=False)
display(efficiency.round(4))
```

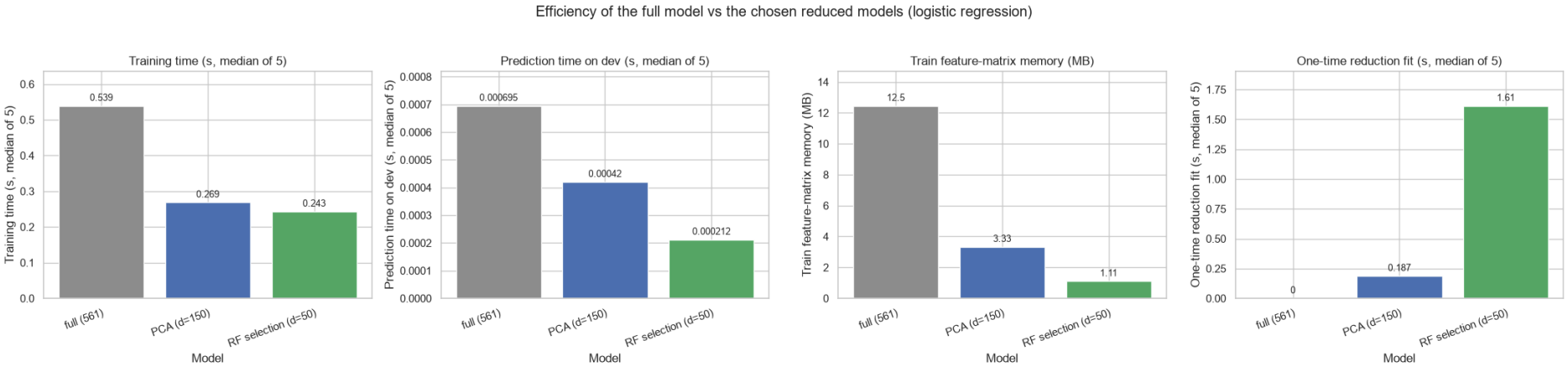
```
saved results\phase5_efficiency.csv
```

| | model | method | d | train_seconds | predict_dev_seconds | reduce_dev_seconds | train_matrix_mb | dev_matrix_mb | one_time_reduction_fit_seconds | train_seconds_vs_full | predict_dev_seconds_vs_full |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | full (561) | full | 561 | 0.5393 | 0.0007 | 0.0000 | 12.4564 | 4.0414 | 0.0000 | 1.0000 | 1.0000 |
| **1** | PCA (d=150) | pca | 150 | 0.2687 | 0.0004 | 0.0039 | 3.3306 | 1.0806 | 0.1874 | 0.4982 | 0.6044 |
| **2** | RF selection (d=50) | rf | 50 | 0.2432 | 0.0002 | 0.0000 | 1.1102 | 0.3602 | 1.6143 | 0.4509 | 0.3051 |

In [41]:
```
panels = [("train_seconds", "Training time (s, median of 5)"),
          ("predict_dev_seconds", "Prediction time on dev (s, median of 5)"),
          ("train_matrix_mb", "Train feature-matrix memory (MB)"),
          ("one_time_reduction_fit_seconds", "One-time reduction fit (s, median of 5)")]
bar_colors = ["#8C8C8C", "#4C72B0", "#55A868"]
fig, axes = plt.subplots(1, 4, figsize=(22, 5))
for ax, (col, label) in zip(axes, panels):
    bars = ax.bar(efficiency["model"], efficiency[col], color=bar_colors)
```

```
    ax.bar_label(bars, labels=[f"{v:.3g}" for v in efficiency[col]], padding=3, fontsize=9)
    ax.set_title(label)
    ax.set_ylabel(label)
    ax.set_xlabel("Model")
    ax.set_ylim(0, max(efficiency[col].max() * 1.18, 1e-9))
    plt.setp(ax.get_xticklabels(), rotation=20, ha="right", rotation_mode="anchor")
fig.suptitle("Efficiency of the full model vs the chosen reduced models (logistic regression)", y=1.03)
fig.tight_layout()
save_fig(fig, "phase5_efficiency")
plt.show()
```

```
saved figures\phase5_efficiency.png
```




## 5.6 FINAL TEST - the only evaluation on the test set

The test arrays are loaded from the cache here for the first time since Phase 1. The full model and the two chosen reduced models are retrained on train (same settings, same d, reducers fit on train) and evaluated once on test.

In [42]:
```
test_data = load_cache()
X_te, y_te_arr = test_data["X_test"], test_data["y_test"]
del test_data
print(f"test: {X_te.shape}")

test_rows, test_reports = [], {}
for name, (m, d) in MODELS.items():
    clf = make_clf().fit(reduce_features(m, d, X_tr), y_tr_arr)
    pred = clf.predict(reduce_features(m, d, X_te))
```

```
    p, r, f, _ = precision_recall_fscore_support(y_te_arr, pred, average="macro", zero_division=0)
    dev_acc = chosen_table.set_index("d").loc[d, "dev_accuracy"]
    test_rows.append({"model": name, "method": m, "d": d, "test_accuracy": accuracy_score(y_te_arr, pred),
                      "test_macro_precision": p, "test_macro_recall": r, "test_macro_f1": f, "dev_accuracy": dev_acc})
    rep = pd.DataFrame(classification_report(y_te_arr, pred, labels=ACTIVITY_ORDER, output_dict=True,
                                             zero_division=0)).T
    rep["support"] = rep["support"].astype(int)
    test_reports[name] = rep
    save_table(rep.round(4), "phase5_test_classification_report_" + name.split(" ")[0].lower())

test_results = pd.DataFrame(test_rows)
test_results["test_acc_gap_vs_full_pp"] = 100 * (test_results["test_accuracy"] - test_results.loc[0, "test_accuracy"])
save_table(test_results, "phase5_test_results", index=False)
display(test_results.round(4))
for name, rep in test_reports.items():
    print(f"\nPer-class classification report on TEST - {name}")
    display(rep.round(3))
```

```
test: (2947, 561)
saved results\phase5_test_classification_report_full.csv
saved results\phase5_test_classification_report_pca.csv
saved results\phase5_test_classification_report_rf.csv
saved results\phase5_test_results.csv
```

| | model | method | d | test_accuracy | test_macro_precision | test_macro_recall | test_macro_f1 | dev_accuracy | test_acc_gap_vs_full_pp |
|---|---|---|---|---|---|---|---|---|---|
| **0** | full (561) | full | 561 | 0.9522 | 0.9543 | 0.9503 | 0.9514 | 0.9572 | 0.0000 |
| **1** | PCA (d=150) | pca | 150 | 0.9423 | 0.9436 | 0.9403 | 0.9413 | 0.9545 | -0.9841 |
| **2** | RF selection (d=50) | rf | 50 | 0.8795 | 0.8868 | 0.8767 | 0.8783 | 0.9473 | -7.2616 |

```
Per-class classification report on TEST - full (561)
```

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **WALKING** | 0.951 | 0.986 | 0.968 | 496 |
| **WALKING_UPSTAIRS** | 0.952 | 0.966 | 0.959 | 471 |
| **WALKING_DOWNSTAIRS** | 0.977 | 0.910 | 0.942 | 420 |
| **SITTING** | 0.960 | 0.876 | 0.916 | 491 |
| **STANDING** | 0.889 | 0.966 | 0.926 | 532 |
| **LAYING** | 0.996 | 0.998 | 0.997 | 537 |
| **accuracy** | 0.952 | 0.952 | 0.952 | 0 |
| **macro avg** | 0.954 | 0.950 | 0.951 | 2947 |
| **weighted avg** | 0.953 | 0.952 | 0.952 | 2947 |

```
Per-class classification report on TEST - PCA (d=150)
```

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **WALKING** | 0.951 | 0.980 | 0.965 | 496 |
| **WALKING_UPSTAIRS** | 0.922 | 0.958 | 0.940 | 471 |
| **WALKING_DOWNSTAIRS** | 0.972 | 0.893 | 0.931 | 420 |
| **SITTING** | 0.929 | 0.874 | 0.900 | 491 |
| **STANDING** | 0.888 | 0.938 | 0.912 | 532 |
| **LAYING** | 1.000 | 1.000 | 1.000 | 537 |
| **accuracy** | 0.942 | 0.942 | 0.942 | 0 |
| **macro avg** | 0.944 | 0.940 | 0.941 | 2947 |
| **weighted avg** | 0.943 | 0.942 | 0.942 | 2947 |

```
Per-class classification report on TEST - RF selection (d=50)
```

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **WALKING** | 0.877 | 0.990 | 0.930 | 496 |
| **WALKING_UPSTAIRS** | 0.848 | 0.924 | 0.884 | 471 |
| **WALKING_DOWNSTAIRS** | 0.991 | 0.800 | 0.885 | 420 |
| **SITTING** | 0.826 | 0.727 | 0.774 | 491 |
| **STANDING** | 0.779 | 0.859 | 0.817 | 532 |
| **LAYING** | 1.000 | 0.961 | 0.980 | 537 |
| **accuracy** | 0.880 | 0.880 | 0.880 | 0 |
| **macro avg** | 0.887 | 0.877 | 0.878 | 2947 |
| **weighted avg** | 0.885 | 0.880 | 0.879 | 2947 |

## 5.7 Summary

In [43]:

```
tr = test_results.set_index("method")
ef = efficiency.set_index("method")
print("=" * 78)
print("PHASE 5 SUMMARY (classifier: LogisticRegression(max_iter=5000), trained on train)")
print("=" * 78)
print(f"Dev baseline, all 561 features: accuracy {100 * base_acc:.2f}%, macro F1 {base_f1:.4f}")
for m, label in [("pca", "PCA"), ("rf", "RF selection")]:
    row = chosen_table.set_index("method").loc[m]
    print(f"Chosen {label:12s}: d = {int(row['d']):3d} -> dev accuracy {100 * row['dev_accuracy']:.2f}% "
          f"({100 * (row['dev_accuracy'] - base_acc):+.2f} pp), within 1 pp: {bool(row['within_1pp_of_full'])}")
print("-" * 78)
print("TEST (single evaluation):")
for m in ["full", "pca", "rf"]:
    r = tr.loc[m]
    print(f"  {r['model']:22s} accuracy {100 * r['test_accuracy']:.2f}%  macro P/R/F1 "
          f"{r['test_macro_precision']:.4f} / {r['test_macro_recall']:.4f} / {r['test_macro_f1']:.4f}  "
          f"({r['test_acc_gap_vs_full_pp']:+.2f} pp vs full)")
```

```
print("-" * 78)
print("Efficiency vs full model:")
for m in ["pca", "rf"]:
    e = ef.loc[m]
    print(f"  {e['model']:22s} features {561} -> {int(e['d'])} ({100 * (1 - e['d'] / 561):.1f}% fewer), "
          f"train time x{e['train_seconds_vs_full']:.2f}, predict time x{e['predict_dev_seconds_vs_full']:.2f}, "
          f"train matrix {e['train_matrix_mb']:.2f} MB vs {ef.loc['full', 'train_matrix_mb']:.2f} MB, "
          f"one-time reducer fit {e['one_time_reduction_fit_seconds']:.2f}s")
print("-" * 78)
print("Most confused pair on dev:", {k: f"{v[0][0][0]} <-> {v[0][0][1]}" for k, v in top_pairs.items()})
```

```
================================================================================
PHASE 5 SUMMARY (classifier: LogisticRegression(max_iter=5000), trained on train)
================================================================================
Dev baseline, all 561 features: accuracy 95.72%, macro F1 0.9534
Chosen PCA         : d = 150 -> dev accuracy 95.45% (-0.28 pp), within 1 pp: True
Chosen RF selection: d =  50 -> dev accuracy 94.73% (-1.00 pp), within 1 pp: True
--------------------------------------------------------------------------------
TEST (single evaluation):
  full (561)             accuracy 95.22%  macro P/R/F1 0.9543 / 0.9503 / 0.9514  (+0.00 pp vs full)
  PCA (d=150)            accuracy 94.23%  macro P/R/F1 0.9436 / 0.9403 / 0.9413  (-0.98 pp vs full)
  RF selection (d=50)    accuracy 87.95%  macro P/R/F1 0.8868 / 0.8767 / 0.8783  (-7.26 pp vs full)
--------------------------------------------------------------------------------
Efficiency vs full model:
  PCA (d=150)            features 561 -> 150 (73.3% fewer), train time x0.50, predict time x0.60, train matrix 3.33 MB vs 12.46 M
B, one-time reducer fit 0.19s
  RF selection (d=50)    features 561 -> 50 (91.1% fewer), train time x0.45, predict time x0.31, train matrix 1.11 MB vs 12.46 MB,
one-time reducer fit 1.61s
--------------------------------------------------------------------------------
Most confused pair on dev: {'full (561)': 'WALKING_UPSTAIRS <-> WALKING_DOWNSTAIRS', 'PCA (d=150)': 'WALKING_UPSTAIRS <-> WALKING_
DOWNSTAIRS', 'RF selection (d=50)': 'WALKING_UPSTAIRS <-> WALKING_DOWNSTAIRS'}
```

## Report - 3. Classification

**Setup.** Every condition uses the same `LogisticRegression(max_iter=5000)`, trained on train and compared on dev. We compared two ways of reducing to d dimensions. **Transform:** the top-d PCA components, which are new features, each a combination of all 561. **Select:** the top-d original features ranked by random-forest impurity importance (300 trees, fit on train).

**Which features the forest selects** ( `phase4_rf_top20_importance.png` , `phase4_rf_cumulative_importance.png` ). Of the top 20 features, 11 are gravity-acceleration features and 3 are angle-to-gravity features. Together they hold about 66% + 17% of the top-20 importance. None are

gyroscope features. These features describe how the phone is oriented, which is what separates laying, sitting and standing, the classes clustering could not split. Importance is spread thinly: 43 features reach 50% of total importance, 137 reach 80% and 232 reach 90%. Many top features are near-duplicates (mean, min, max and energy of the same gravity axis), which dilutes the credit across them.

**Performance vs number of dimensions** ( `phase5_dev_accuracy_vs_d.png` ). The full 561-feature model reaches **95.7%** dev accuracy (macro F1 0.953).

- **At small d, PCA is far stronger.** At d = 9 PCA reaches 87.6% while selection reaches only 66.6%. The elbow value k = 3 gives just 71.5% with PCA, so the elbow keeps too little for classification. PCA components are uncorrelated by construction, while the top RF features are redundant copies of the same gravity signals, so a handful of them carries little new information.
- **From d ≈ 20–30 upward, selection catches up and then overtakes PCA.** It peaks at 96.1% at d = 75, slightly above the full baseline, and PCA reaches the baseline only around d = 150–200.

We chose, for each approach, the smallest d whose dev accuracy is within 1 percentage point of the full model: **PCA d = 150** (95.45%, −0.28 pp) and **RF selection d = 50** (94.73%, −1.00 pp, just inside the threshold).

**Confusion on dev** ( `phase5_dev_confusion_matrices.png` ). All three models make the same three main mistakes:

- walking downstairs predicted as walking upstairs (14–19% of downstairs windows)
- standing predicted as sitting (5–10%)
- walking downstairs predicted as walking (3–5%)

Laying is near-perfect in all three, and errors between moving and stationary activities are almost absent (0.1–0.2% of dev windows), consistent with the clustering results. Reducing the features mainly worsens the errors that already existed: the RF model's standing→sitting confusion rises to 10% and its downstairs→upstairs confusion to 19%.

**Final test** (Section 5.6, the only time a model is trained or evaluated with test data):

| Model | Test accuracy | Macro precision | Macro recall | Macro F1 |
|---|---|---|---|---|
| Full (561) | 95.22% | 0.954 | 0.950 | 0.951 |
| PCA (d = 150) | 94.23% | 0.944 | 0.940 | 0.941 |
| RF selection (d = 50) | 87.95% | 0.887 | 0.877 | 0.878 |

PCA generalizes as predicted, about 1 point below the full model. RF selection does not: it loses 7.3 points on test despite being within 1 point on dev. Its sitting recall falls to 0.73 (full model 0.88) and its standing recall to 0.86 (full model 0.97). These are exactly the classes the selected gravity and orientation features are responsible for. The likely cause is that phone orientation varies from person to person, and d = 50 was chosen right

at the threshold using a dev set of only 5 subjects. The large gap between dev (−1.0 pp) and test (−7.3 pp) suggests that this dev set was too small to pick d reliably for the selection approach. We did not re-select d after seeing the test results, since that would contaminate the test set.

**Interpretability.** Selection keeps original, named features, so we can say the model relies on phone orientation. PCA components mix all 561 features and can only be interpreted loosely through their top loadings.

**Efficiency** ( `phase5_efficiency.png` ). Compared with the full model:

- **PCA (d = 150):** training time ×0.50, train matrix 3.3 MB instead of 12.5 MB, one-time PCA fit ≈ 0.19 s.
- **RF selection (d = 50):** training time ×0.45, prediction time ×0.31, train matrix 1.1 MB, one-time forest fit ≈ 1.6 s.

Prediction times are below a millisecond for the whole dev set, so the ratios vary between runs.

The more important difference is one these timings don't show. PCA still needs all 561 features to be computed for every new window before it can be projected, while selection needs only 50 to be computed at all. On a phone, feature extraction from raw signals is likely to cost more than the classifier itself. Selection could therefore save far more in deployment than the training-time numbers suggest, but in our test it paid for that in accuracy on unseen people.

**How the two approaches compare.** PCA is the safer reduction: it degrades gracefully, is strong even at small d, and generalized from dev to test. Selection is more interpretable and potentially cheaper to deploy, and at moderate d it can match or beat the full model on dev, but it concentrated the model on person-dependent orientation features and generalized worse.

# Report

Reflection on the lab and the statement on generative-AI use.

## 4. Reflection

The most challenging part was choosing hyperparameters for the unsupervised methods without using the labels, especially DBSCAN's eps. A single eps value that looked reasonable in one feature space was meaningless in another: the knee was 4.1 in three PCA dimensions and 21.8 in 561. We handled this with a label-free rule, the highest silhouette under a noise cap, and an explicit sweep, and only used the labels afterwards to evaluate. The main takeaway is that how the data are split matters as much as the model. Splitting dev by subject, rather than by row, gave honest estimates: the full model's dev accuracy (95.7%) matched its test accuracy (95.2%). The surprise was the RF-selected model, which looked as good

as the full model on dev but lost 7 points on test, because the features it depended on vary across people. Feature reduction can look free on a small validation set and still cost a lot on new users.

## 5. Generative AI Use Statement (Option B)

| Phase | Used AI? | How (mode + what you asked) | How you checked / corrected / overrode it |
|---|---|---|---|
| Planning & understanding | Yes | **Information / Advice** (Claude chat): asked how to approach the lab, how to split dev without leakage (by subject), how to pick the PCA elbow, and what eps/silhouette mean for DBSCAN. | Read each spec against the assignment before running it. Asked follow-up questions when something was unclear (e.g. why k = 3 and not a later elbow, why DBSCAN ran on PCA components) and added checks where the answer depended on an arbitrary choice (elbow robustness across 30/50/100 components, a 561-D DBSCAN comparison). |
| Coding and Debugging | Yes | **Creation** (Claude Code in VS Code): generated all notebook code for Phases 1–5, including the split, scaling, PCA, K-Means, DBSCAN sweeps, RF ranking, the d sweep, timing and the final test. | Reviewed outputs after every phase before allowing the next one. Asked for fixes: overlapping axis labels, an unreadable k-distance axis, and a divide-by-near-zero timing column that was removed. Verified that the test set is used only in Section 5.6 and corrected an overstated claim about this in the notebook text. |
| Analysis & interpretation | Yes | **Advice / feedback** (Claude chat): asked Claude to check each phase's figures and CSVs and explain what they showed (e.g. why PCA beats selection at small d, why RF selection dropped on test, why eps differs between 3-D and 561-D). | Cross-checked the explanations against the actual numbers in `results/`, kept only claims the data supported, and rejected re-selecting d after seeing test results. |
| Report writing, formatting, editing | Yes | **Creation:** Claude drafted the report text from my results. | Reviewed every section and checked all numbers against the CSVs in results/; made some wording edits, including in the reflection, which Claude drafted. |

AI tools used: Claude (Anthropic) in chat, and Claude Code in VS Code.